

ANSWERS & EXPLANATIONS
GENERAL STUDIES (P) TEST – 6337

- Q 1. A** • Coral reefs are known as the "Rainforests of the Ocean". This comparison is drawn because, much like terrestrial tropical rainforests, coral reefs host an extraordinary level of biodiversity and exhibit immense primary productivity, despite covering less than 1% of the ocean floor. **Hence, statement I is correct.**
- Tropical waters are typically oligotrophic (nutrient-poor), meaning they lack significant amounts of dissolved nitrogen and phosphorus.
 - Despite this, reefs thrive because of efficient internal recycling. The best example of this is the symbiotic relationship between coral polyps and Zooxanthellae (algae). The waste products of the coral (CO₂ and nitrogenous waste) are used by the algae for photosynthesis, and the algae, in turn, provide oxygen and organic nutrients back to the coral. **Hence, statement II is correct.**
 - Statement-II explains why Statement-I is true. The "Rainforest" label is given precisely because these ecosystems manage to produce a massive amount of life and energy in an environment that would otherwise be a biological desert. **Hence, option (a) is the correct answer.**
- Q 2. B** Black carbon is a short-lived climate pollutant formed by the incomplete combustion of fossil fuels, biofuels, and biomass. The major contributors are categorized as follows:
- Diesel engines are one of the primary global sources of black carbon. Because diesel engines operate under high pressure and temperature with heterogeneous fuel-air mixing, they inherently produce significant amounts of particulate matter (soot), which is largely composed of black carbon. **Hence, point 1 is correct.**
- Open biomass burning (including crop residue burning after harvests and wildfires) is a massive source of black carbon. This is especially prevalent in developing regions like South Asia and Southeast Asia, contributing significantly to regional "Atmospheric Brown Clouds." **Hence, point 2 is correct.**
- While coal combustion is a potential source, large-scale thermal power plants are generally not major contributors to black carbon compared to the other sources. Specifically, Electrostatic Precipitators (ESPs) are highly efficient at capturing fine particulate matter (PM_{2.5}), including black carbon, before it is emitted from the stack. Consequently, their contribution is significantly lower than that of unregulated open burning or inefficient small-scale combustion. **Hence, point 3 is not correct.**
- Traditional biomass burning for cooking and heating is the largest single source of black carbon globally. The use of traditional "chulhas" or open fires using wood, dung, and charcoal for residential energy involves incomplete combustion, releasing vast amounts of black carbon directly into the domestic and local environment. **Hence, point 4 is correct.**
- Q 3. A** • GNP measures the total monetary value of the output produced by a country's residents. Therefore, any output produced by foreign residents within the country's borders must be excluded in calculations of GNP, while any output produced by the country's residents outside of its borders must be counted. **Hence statement 1 is not correct.**

- **GNP does not include intermediate goods and services** to avoid double-counting since they are already incorporated in the value of final goods and services. **Hence statement 2 is correct.**
- GNP and GDP are very closely related concepts, and the main differences between them come from the fact that there may be companies owned by foreign residents that produce goods in the country, and companies owned by domestic residents that produce goods for the rest of the world and revert earned income to domestic residents.
- **Income from overseas investments by a country's residents counts in GNP, and foreign investment within a country's borders does not.** This is in contrast to GDP which measures economic output and income based on location rather than nationality. In other words, the Gross Domestic Product (GDP) measures economic output based on location/territory (production within borders), while Gross National Product (GNP) measures output based on citizenship/residency (production by residents, regardless of where they are located).
- While GDP is the most widely followed measure of a country's economic activity, GNP is still worth looking at because large differences between GNP and GDP may indicate that a country is becoming more engaged in international trade, production, or financial operations. The larger the difference between a country's GNP and GDP, the greater the degree of incomes and investment activity in that country involve transnational activities such as foreign direct investment one way or another.

Q 4. A • India and France started their diplomatic relationship in the year 1947.

- Recently, in February 2026, India and France officially elevated their bilateral relationship to a "Special Global Strategic Partnership." This landmark upgrade, announced during French President Emmanuel Macron's fourth state visit to India, signifies a transition from a sector-specific "strategic partnership" (established in 1998) to a comprehensive, long-term alliance aimed at global stability and joint innovation. **Hence option (a) is the correct answer.**

Q 5. C • **Consumer Price Index(CPI) the index of prices of a given basket of commodities that are bought by the representative consumer.** CPI is generally expressed in percentage terms. We have two years under consideration – one is the base year, the other is the current year. We calculate the cost of purchase of a given basket of commodities in the base year.

- The **Laspeyres Index is the primary formula used to calculate the Consumer Price Index (CPI).** It measures the change in the cost of a fixed basket of goods and services over time.
- **CPI (Rural/Urban/Combined) - "Headline Inflation" Releasing Body: National Statistical Office (NSO), Ministry of Statistics and Programme Implementation (MoSPI).**

This is the most important index used by the Reserve Bank of India (RBI) for inflation targeting and monetary policy. It measures retail inflation for all-India and state/UT levels.

- **Base Year: 2012=100 (Earlier), with a new series on Base 2024=100 launched in February 2026.**

CPI for Industrial Workers (CPI-IW) Releasing Body: Labour Bureau, Ministry of Labour and Employment.

- This index is used to determine the Dearness Allowance (DA) for central/state government employees and workers in key industrial sectors. **Base Year: 2016=100. Labour Bureau Labour Bureau.**

CPI for Agricultural Labourers (CPI-AL) & Rural Labourers (CPI-RL) Releasing Body: Labour Bureau, Ministry of Labour and Employment.

- These are used to determine minimum wages for agricultural and rural labourers, respectively, across different states.
 - Base Year: Updated to 2019=100 effective from June 2025.
- Consumer Food Price Index (CFPI) Releasing Body: National Statistical Office (NSO), MoSPI.
Details: Measures the change in retail prices of food products specifically.
- It is a sub-index of the overall CPI-Combined.
- **Hence only pairs 1 and 2 are correctly matched.**

Q 6. C • The Monopolies and Restrictive Trade Practices Act, 1969 (MRTP Act) was repealed and replaced by the Competition Act, 2002, on the recommendations of Raghavan committee. Hence statement 1 is correct.

The Competition Act was passed in 2002 and has been amended by the Competition (Amendment) Act, 2007. It follows the philosophy of modern competition laws.

- The Act prohibits **anti-competitive agreements**, abuse of dominant position by enterprises and regulates combinations (acquisition, acquiring of control and M&A), which causes or likely to cause an appreciable adverse effect on competition within India.
 - **The Competition Commission of India (CCI) can penalise companies involved in anti-competitive agreements.** For example, CCI imposed a fine of ₹258 crores upon Three Airlines in 2015. CCI had penalized the three airlines for cartelisation in determining the fuel surcharge on air cargo. **Hence statement 3 is correct.**
- In accordance with the provisions of the Amendment Act, the Competition Commission of India has been established.
- The commission is a quasi-judicial body which gives opinions to statutory authorities and also deals with other cases.
 - The Commission consists of one Chairperson and six Members as per the Competition Act who shall be appointed by the Central Government.
- **The National Company Law Appellate Tribunal (NCLAT) was constituted under section 410 of Companies Act, 2013.** However, as per section 53A of Competition Act, 2002, the NCLAT will function as the Appellate Tribunals for the purposes of the Competition Act, 2002 as well. **Hence statement 2 is not correct.**

Q 7. B Food Safety and Standards Authority of India (FSSAI) mandated the display of chicory content in front of the coffee pack.

About Chicory

- It is a blue-flowered perennial plant of the family Asteraceae.
- Addition to Coffee: Its root can be roasted and grounded to impart additional colour, and bitterness to coffee, caffeine-free serves as a substitute for coffee, Contains inulin which is a natural sweetener and a probiotic.
- FSSAI regulations (2011) mandate not less than 51% coffee content in coffee-chicory mixture.

Hence, option (b) is the correct answer.

- Q 8. A • A Non-Banking Financial Company (NBFC) is a company registered under the Companies Act, 1956/2013 engaged in the business of loans and advances, acquisition of shares/stocks/bonds/debentures/securities issued by Government or local authority or other marketable securities of a like nature, leasing, hire-purchase, insurance business, chit business but does not include any institution whose principal business is that of agriculture activity, industrial activity, purchase or sale of any goods (other than securities) or providing any services and sale/purchase/construction of immovable property. Hence statement 1 is not correct.**

NBFCs lend and make investments and hence their activities are similar to that of banks; however there are a few differences as given below:

- NBFC cannot accept demand deposits.
- **NBFCs do not form part of the payment and settlement system and cannot issue cheques drawn on itself. Hence statement 2 is correct.**
- Deposit insurance facility of Deposit Insurance and Credit Guarantee Corporation is not available to depositors of NBFCs, unlike in case of banks.
- Housing Finance Companies, Merchant Banking Companies, Stock Exchanges, Companies engaged in the business of stock-broking/sub-broking, Venture Capital Fund Companies, Nidhi Companies, Insurance companies and Chit Fund Companies are NBFCs but they have been exempted from the requirement of registration under the RBI Act, 1934 subject to certain conditions.
- **They are regulated by different institutions depending on their activity such Housing Finance Co. regulated by National Housing Bank, Merchant Banking Companies and Mutual Funds etc by SEBI, Insurance Co. by IRDA, Asset Finance company by RBI, etc. Hence statement 3 is not correct.**

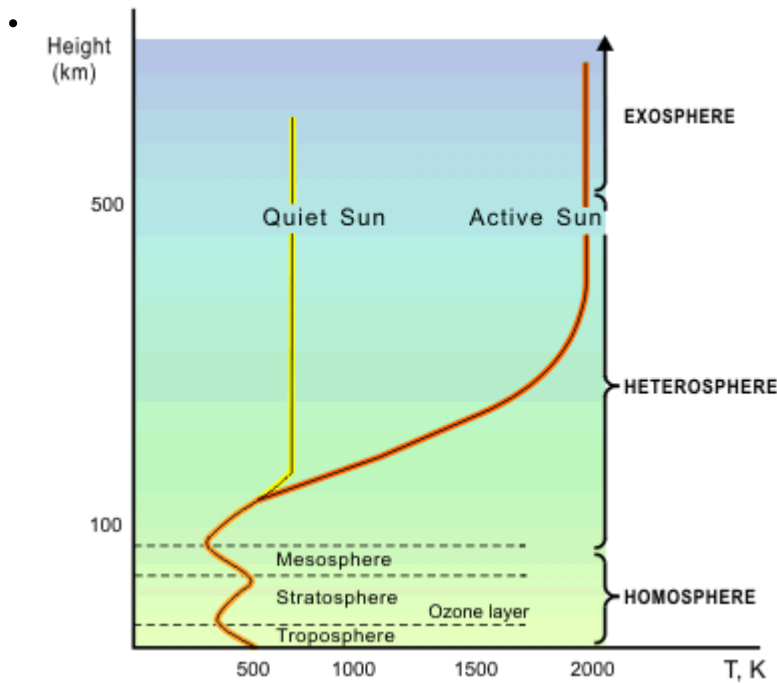
- Q 9. B • Vaccines are preventive biological preparations designed to prepare the immune system in advance against specific diseases.**

They work by introducing into the body:

- Weakened (attenuated) pathogens
- Inactivated (killed) pathogens
- Specific antigenic components (such as proteins)
- Genetic material coding for antigens (in some modern vaccines)
- These components do not cause the actual disease, but they trigger the immune system to:
 1. Produce antibodies
 2. Activate B-cells and T-cells
 3. Form memory cells
- Because of this immunological memory, if the real pathogen later enters the body, the immune system responds rapidly and effectively, either preventing infection or significantly reducing disease severity.
- Thus, vaccines stimulate the body's immune system to develop protection. **Hence, option (b) is the correct answer.**
- **Other options:**
- (a) Vaccines are preventive, not curative. Neutralizing toxins after disease onset is the role of antitoxins or therapeutic antibodies.

- (c) Vaccines do not directly kill pathogens. Drugs like antibiotics or antivirals may kill or inhibit pathogens. Vaccines instead train the immune system to fight future infections.
- (d) Vaccines are effective against both bacterial and viral diseases. Examples include protection against viral infections such as measles and polio, as well as bacterial infections like tetanus.

Q 10. D • The Homosphere is the layer of the atmosphere (extending from the surface to about 80 km) where the major gaseous constituents (Nitrogen, Oxygen, and Argon) are mixed uniformly. The Ozone Layer (Ozonosphere) is primarily located in the Stratosphere, between 15 km and 35 km above the Earth's surface. Since 35 km is well within the 80 km limit of the Homosphere, the the ozone layer lies entirely within the Homosphere is geographically and chemically correct. **Hence, statement 1 is correct.**



- Water vapour is a highly variable gas that decreases rapidly with altitude.
- Because it is sourced from surface evaporation and is relatively "heavy" compared to other trace gases, nearly 90% of the total water vapour in the atmosphere is found within the first 6 km (the lower Troposphere). As the air cools with height, its capacity to hold moisture drops, leading to this sharp decrease. **Hence, statement 2 is correct.**
- In a dry atmosphere, the proportions of gases by volume are strictly ranked. Argon is the third most abundant gas, making up about 0.93%. Carbon dioxide is a trace gas making up about 0.04% (approximately 410–420 ppm).
- Therefore, the proportion of Argon is significantly higher than that of Carbon dioxide. **Hence, statement 3 is correct.**

Q 11. C • To mark the 40th Foundation Day, the Department of Biotechnology (DBT) launched SUJVIKA.

About SUJVIKA

- An AI-driven Biotech Product Data Portal, developed in collaboration with Industry partners ABLE.

- It serves as a Trade Statistics Digital Intelligence Platform, providing authenticated biotechnology product import data in a structured and accessible format.
- The portal provides sector-wise insights into biochemical products, industrial enzymes, and other biotechnology imports.
- It enables researchers and startups to identify high-value imports, assess dependencies, and prioritise indigenous R&D.

• **Hence, option (d) is the correct answer.**

•

Q 12. D • The evolution of Earth's atmosphere from a reducing (oxygen-poor) state to an oxidizing (oxygen-rich) state is one of the most significant events in planetary history.

- The Great Oxidation Event (GOE), occurring approximately 2.4–2.1 billion years ago, is most closely associated with the massive biological production of oxygen by cyanobacteria as a byproduct of photosynthesis. This accumulation of oxygen in the atmosphere significantly transformed Earth's environment.

- **The Biological Mechanism:** Around 2.4 billion years ago, cyanobacteria (blue-green algae) became the first organisms to develop oxygenic photosynthesis. By using sunlight to split water molecules, they released free oxygen as a waste product.

- **The Atmospheric Shift:** Initially, this oxygen was absorbed by "sinks" like dissolved iron in the oceans, forming Banded Iron Formations (BIFs). Once these sinks were saturated, oxygen began accumulating in the atmosphere, transforming it from a reducing environment (rich in methane and ammonia) to an oxidizing one.

• **Hence, option (d) is the correct answer.**

- While volcanoes release many gases (like CO₂ and H₂O), they do not release free oxygen. The breakup of Rodinia occurred much later, around 750 million years ago.

Q 13. D • **The Vijayanagara empire which was founded by two brothers Harihara Raya-I and Bukka Raya-I in 1336 between river Krishna and Tungabhadra, reached its golden age under the reign of the powerful king, Sri Krishnadevaraya.**

- **Krishnadeva Raya was known as Abhinava Bhoja. Eight great Telugu poets adorned his court.** They were known as Ashtadiggajas. **Allasani Peddanna**, Nandi Thimmanna, **Tenali Ramakrishna**, Dhurjathy, Bhattu Murthy, Puna Vira Bhadra, Mallana and Pingali Suranna.

- **Allasani Peddanna** was the greatest and he was often **described as 'Andhrakavita Pitamaha'.** Allasani Peddanna was the author of Manucharita in Telugu. **Tenali Rama was the most famous scholar and jester of Krishnadeva Raya's court.** **Krishnadeva Raya himself wrote Amuktamlayada in Telugu.** It relates the story of the daughter of Periazavar, Goda Devi (Andal). He was himself a scholar of Sanskrit as well as Telugu.

• **Hence, option (d) is the correct answer.**

Q 14. D LiDAR (Light Detection and Ranging) is a versatile remote sensing technology that uses laser pulses to measure distances and create precise 3D models. It is effectively used in following applications:

- Creation of high-resolution topographic maps: LiDAR is a primary tool for generating detailed 3D representations of terrain, including Digital Elevation Models (DEMs) and Digital Terrain Models (DTMs).
- Detection of underwater mineral deposits using light waves: Bathymetric LiDAR uses green light (which can penetrate water) to map seafloor topography and identify underwater features, supporting mineral exploration and nautical charting.
- Autonomous vehicle navigation: It provides critical real-time 3D vision, allowing self-driving cars to detect and avoid obstacles, recognize lane markings, and navigate complex environments safely.
- Forest canopy height and biomass assessment: In forestry, LiDAR is used to measure tree heights, canopy density, and volume to estimate biomass and carbon stocks.
- Weather forecasting through cloud profiling: Atmospheric LiDAR systems measure cloud height, density, and aerosol distribution, providing essential data for meteorological models and weather predictions.

Hence, option (d) is the correct answer.

- Q 15. D** Trachoma is not a viral infection of the respiratory tract. It is a chronic infectious bacterial eye disease caused by *Chlamydia trachomatis*. It affects the conjunctiva and can lead to blindness, rather than affecting the lungs or spreading primarily through airborne droplets (it spreads through contact with eye/nose discharge, often via flies or contaminated items). It is categorised as a Neglected Tropical Disease by the WHO. **Hence, statement 1 is not correct.**

India was awarded the Certificate of Elimination of Trachoma as a Public Health Problem by the WHO. India was declared the 3rd (after Nepal and Myanmar) Southeast Asian country to have eliminated Trachoma. **Hence, statement 2 is not correct.**

Causes & Transmission

- Pathogen: *Chlamydia trachomatis* bacteria.
- Transmission: Direct contact with eye/nose discharge from infected individuals (especially children), or indirectly via towels, bedding, and flies.
- Reservoir: Preschool-age children often act as the primary reservoir for infection

- Q 16. D** According to a report by Greenpeace (an environmental Non-Governmental Organization), India is the largest emitter of Sulphur Dioxide (SO₂) in the world, contributing more than 15% of global anthropogenic emissions. **Hence, statement 1 is correct.**

The primary sources of in India are coal-based thermal power plants and industrial processes (such as smelting). India's heavy reliance on coal for electricity generation, combined with a lack of flue-gas desulfurization (FGD) technology in many plants, makes these sectors the largest contributors. **Hence, statement 2 is correct.**

Sulphur dioxide is one of the eight major pollutants monitored and reported under India's National Air Quality Index (AQI). The full list of monitored pollutants includes:

- Particulate Matter
- Particulate Matter
- Nitrogen Dioxide

- **Sulphur Dioxide Hence, statement 3 is correct.**
- Carbon Monoxide
- Ozone
- Ammonia
- Lead

- Q 17. A**
- A keystone species is defined by its disproportionately large effect on its environment relative to its abundance. Removing such a species often leads to trophic cascades, significant alterations in community structure, or the complete collapse of the ecosystem. **Hence, statement 1 is correct.**
 - While many apex predators (like wolves or sea stars) are keystone species, the two terms are not synonymous. An apex predator is defined by its position at the top of the food chain, whereas a keystone species is defined by its functional impact. Not all top predators exert the unique, system-wide control required to be classified as a keystone. Conversely, many keystone species are not predators at all, such as ecosystem engineers (beavers) or mutualists (bees). **Hence, statement 2 is not correct.**
 - Keystone species are not required to be the most abundant; rather, they are defined by having a disproportionately large, critical impact on their ecosystem relative to their low abundance or biomass. They often hold ecosystems together, and their removal causes drastic, negative changes, but they are rarely the most numerous. **Hence, statement 3 is not correct.**

- Q 18. B** According to the Wild Life (Protection) Act, 1972, any diversion of land or alteration of boundaries in certain protected areas requires the recommendation or approval of the National Board for Wildlife (NBWL).

- **National Parks:** Any alteration of boundaries or diversion of land within a National Park requires the recommendation of the NBWL. **Hence, point 1 is correct.**
- **Wildlife Sanctuaries:** Similar to National Parks, no alteration of boundaries or diversion of land in a Wildlife Sanctuary can be done without the approval/recommendation of the NBWL. **Hence, point 2 is correct.**
- **Conservation Reserves:** These are a category of protected areas introduced under the Wild Life (Protection) Amendment Act, 2002. They are also subject to the regulatory oversight of the NBWL regarding land diversion and boundary changes. **Hence, point 3 is correct.**

Eco-Sensitive Zones (ESZs): These areas are notified under the Environment (Protection) Act, 1986, rather than the Wild Life (Protection) Act. While projects within ESZs often require "Wildlife Clearance" from the Standing Committee of the NBWL (especially if they require Environmental Clearance), the primary statutory requirement for the diversion of land managed by the NBWL applies specifically to the protected areas defined under the Wild Life (Protection) Act. **Hence, point 4 is not correct.**

- Q 19. A** Ecological succession is the gradual, directional, and predictable process of change in the species structure of an ecological community over time. It begins with pioneer species colonizing a bare area, advancing through intermediate stages to a stable climax community.

While lichens are the most common pioneer species in Xerarch succession (on bare rock), primary succession can also occur in water bodies (Hydrosere). In aquatic primary succession, the pioneer species are usually phytoplankton, not lichens. **Hence, point 1 is not correct.**

Primary succession starts on bare surfaces (like lava or bare rock) where no soil exists, requiring hundreds of years for soil formation. Secondary succession occurs in areas where a previous community was destroyed (e.g., by fire or abandoned farmland) but the soil and nutrients remain, allowing life to return much more quickly. **Hence, point 2 is correct.**

In a climax community, the ecosystem is in a state of equilibrium. While the Net Community Productivity (NCP)—the balance between total photosynthesis and total respiration of the entire community—approaches zero, the Net Primary Productivity (NPP) of the plants themselves is not zero. Plants continue to grow and produce biomass; it is just that the rate of production is roughly equal to the rate of consumption and decomposition within the stable community. **Hence, point 3 is not correct.**

- Q 20. D**
- **Regional Rural Banks were set up under the Regional Rural Banks Act, 1976** with a view to developing the rural economy by providing credit and other facilities, particularly to the small and marginal farmers, agricultural labourers, artisans and small entrepreneurs.
 - The function of **financial regulation over RRBs is exercised by Reserve Bank and the supervisory powers have been vested with NABARD. Hence statement 1 is not correct.**
 - **The equity of the RRBs was contributed by the Central Government, concerned State Government and the sponsor bank in the proportion of 50:15:35. Hence statement 2 is correct.**
 - **CRAR norms or Capital Adequacy Ratio (CAR)** is the ratio of a bank's capital in relation to its risk-weighted assets and current liabilities. It is decided by central banks. **Capital adequacy norms were prescribed for Regional Rural Banks (RRBs) by RBI in its circular dated December 28, 2007, to compute capital to risk-weighted assets (CRAR).** Further, the income recognition, asset classification, and provisioning norms as applicable to commercial banks are applicable to RRBs. The RBI issued the "Regional Rural Banks – **Prudential Norms on Capital Adequacy Directions, 2025**" which define the composition of Tier 1 and Tier 2 capital and mandate strict deductions for losses, goodwill, and certain deferred tax assets. RRBs must maintain a minimum Capital to Risk-Weighted Assets Ratio (CRAR) of 9%. **Hence statement 3 is not correct.**

- Q 21. A**
- Alternative Investment Fund or AIF means any fund established or incorporated in India which is a privately pooled investment vehicle that collects funds from sophisticated investors, whether Indian or foreign, for investing it in accordance with a defined investment policy for the benefit of its investors.
 - In India, applicants can seek registration as an AIF in one of the following categories, and in sub-categories thereof, as may be applicable:

Category I: AIFs which invest in start-up or early-stage ventures or social ventures or SMEs or infrastructure or other sectors or areas which the government or regulators consider as socially or economically desirable.

- **Venture capital funds (Including Angel Funds)**
- **SME Funds**
- **Social Venture Funds**
- **Infrastructure funds**

Category II: AIFs which do not fall in Category I and III and which do not undertake leverage or borrowing other than to meet day-to-day operational requirements and as permitted in the SEBI (Alternative Investment Funds) Regulations.

- Various types of funds such as **real estate funds, private equity funds, funds for distressed assets, etc.** are registered as Category II AIFs.

Category III: AIFs which employ diverse or complex trading strategies and may employ leverage including through investment in listed or unlisted derivatives.

- Various types of funds such as **hedge funds, PIPE Funds, etc.** are registered as Category III AIFs.
- **AIF does not include funds covered under the SEBI (Mutual Funds) Regulations, 1996, SEBI (Collective Investment Schemes) Regulations, 1999 or any other regulations of the Board to regulate fund management activities**
- **Hence, option (a) is the correct answer.**

Q 22. C Lake-Effect Snow:

- Lake-effect snow is a meteorological phenomenon that occurs **when cold air passes over large, relatively warm water bodies, leading to intense, localized snowfall. Hence statement 1 is correct.**
- **This event is most commonly observed near large lakes such as the Great Lakes in North America, particularly during the late fall and winter seasons. Hence statement 2 is also correct.**
- Recently, Lake-Effect Snow has been in the news due to severe snowstorms **in parts of the United States and Canada**, where such occurrences have caused travel disruptions, power outages, and record-breaking snowfall levels.

What is Lake-Effect Snow?

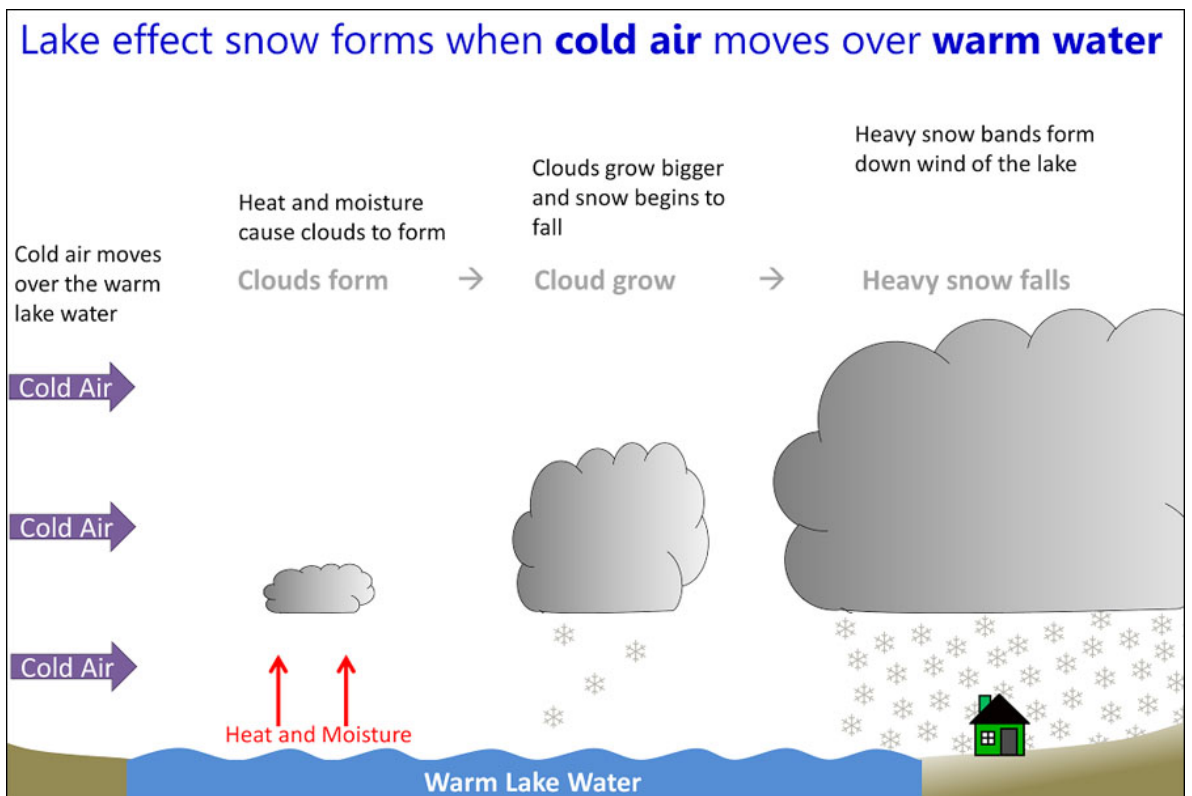
- Lake-effect snow **occurs when cold, dry air masses, typically from Arctic regions**, move over large, unfrozen lakes. As the cold air passes over the warmer water surface, the following process occurs:

Step-by-Step Formation Process

- 1. Heat and Moisture Transfer – The warm lake water warms and humidifies the cold air, increasing its moisture content.
- 2. Rising Air and Cloud Formation – The now warmer and more humid air rises, condenses, and forms cumulus and cumulonimbus clouds.
- 3. Snowfall Development – As the moist air cools rapidly, it produces narrow bands of heavy snowfall on the leeward (downwind) side of the lake.
- 4. Localized Impact – The snowfall is often highly concentrated, meaning one area can receive intense snowfall while a nearby area remains clear.

Key Features of Lake-Effect Snow

- Occurs in late fall and winter when the lake is still warm compared to the overlying air.
- **Commonly observed near large lakes, especially in North America, Japan, and Russia.**
- Localized and intense, with snowfall rates of 2-3 inches per hour or more.
- Heavily dependent on wind direction, meaning some areas receive significantly more snow than others just a few miles away.
- Can cause blizzards, reduced visibility, and hazardous road conditions.



- Q 23. B • Recently, the Government of India announced Rashtriya Vigyan Puraskar 2025.** Rashtriya Vigyan Puraskar is India's highest recognition for outstanding and inspiring contributions by scientists, technologists, and innovators in diverse fields of science, technology, and technology-led innovation.
- **The Rashtriya Vigyan Puraskar is given under four categories:**
 - **Vigyan Ratna (VR) award** - Given for **lifetime achievements & contributions made in any field of science and technology**. Prof. Jayant Vishnu Narlikar has been honoured with the Vigyan Ratna award posthumously. **Hence, pair 3 is correctly matched.**
 - **Vigyan Shri (VS) award** - Given for distinguished contributions in any field of science and technology. **Hence, pair 2 is correctly matched.**
 - **Vigyan Yuva-Shanti Swarup Bhatnagar (VY-SSB) award** - Given to young scientists up to the age of 45 years who have made an exceptional contribution in any field of science and technology. **Hence, pair 1 is not correctly matched.**
 - **Vigyan Team (VT) award** - Given to a team of three or more scientists/researchers/innovators for exceptional teamwork in any field of science and technology. **Vigyan Team award was given to Team Aroma Mission CSIR.**
 - These Awards are conferred across 13 domains: Physics, Chemistry, Biological Sciences, Mathematics & Computer Science, Earth Science, Medicine, Engineering Sciences, Agricultural Science, Environmental Science, Technology & Innovation, Atomic Energy, Space Science and Technology, and other allied fields.

Q 24. C Understanding the Given details:

1. High-altitude wetland and breeding ground for Black-necked Crane:
 - The Black-necked Crane (*Grus nigricollis*) is found in high-altitude wetlands of the Tibetan Plateau and Ladakh region.

- It is critically dependent on specific wetlands for breeding, particularly in Tso Kar Wetland Complex in Ladakh.
- 2. Extreme cold desert conditions with freezing temperatures for most of the year:
 - The Ladakh region experiences harsh winter conditions with sub-zero temperatures for most of the year.
 - Tso Kar Wetland, being in Ladakh, falls under these cold desert conditions.
- 3. Located in a rain-shadow region, making it one of the driest wetlands in India:
 - Ladakh lies in the rain-shadow area of the Himalayas, meaning very little precipitation occurs.
 - This makes Tso Kar one of the driest Ramsar wetlands in India.
- **Analysis of other options:**
 - (a) Loktak Lake (Manipur)
 - It is not a high-altitude wetland but rather a low-altitude freshwater lake.
 - Located in Northeast India, which receives high rainfall, making it not a dry wetland.
 - It is famous for its phumdis (floating biomass islands) and Keibul Lamjao National Park (home to the endangered Sangai deer).
 - (b) Wular Lake (Jammu & Kashmir)
 - It is one of the largest freshwater lakes in India, located in the Kashmir Valley.
 - The lake is not in an extreme cold desert region.
 - Not a primary breeding ground for the Black-necked Crane.
 - (d) Chandra Taal Wetland (Himachal Pradesh)
 - It is a high-altitude wetland in Spiti Valley, Himachal Pradesh.
 - However, it is not a significant breeding ground for the Black-necked Crane.
 - It is smaller and more of a glacial-fed lake than a major wetland complex like Tso Kar.

About Tso Kar Wetland Complex

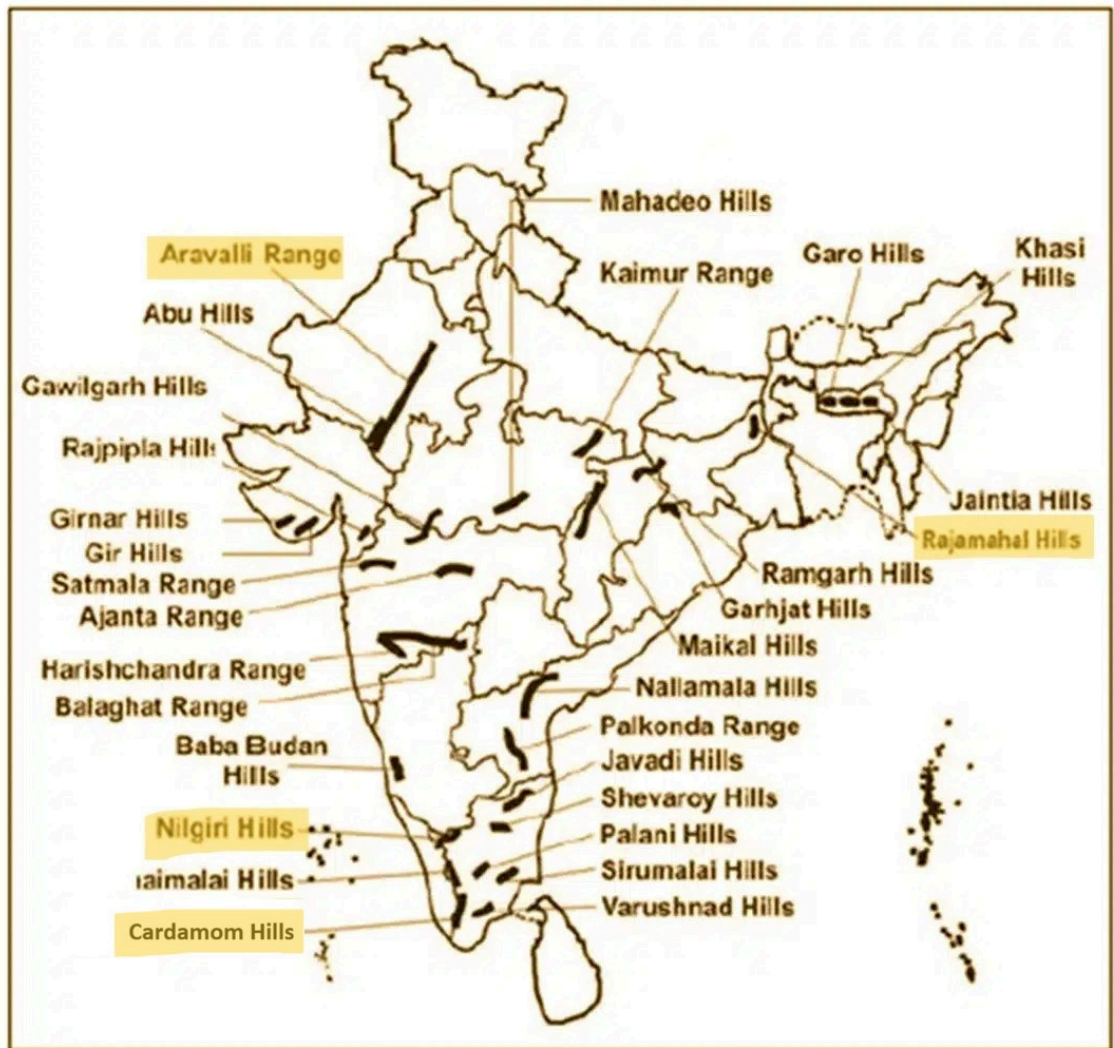
- Location: Ladakh, India
- Declared as Ramsar Site: December 2020
- Type: High-altitude wetland, saltwater lake
- Key Biodiversity:
 - A major breeding ground for the Black-necked Crane (*Grus nigricollis*), an IUCN Vulnerable species.
 - Other species found: Bar-headed geese, brown-headed gulls, and Great crested grebes.

- Q 25. A**
- Most of the ranges in Northeast India are separated from each other by numerous small rivers. The Barak is an important river in Manipur and Mizoram. **Hence statement 3 is correct.**
 - The physiography of Manipur is unique by the presence of a large lake known as 'Loktak' lake at the center, surrounded by mountains from all sides. **Hence statement 1 is not correct.**
 - **Mizoram which is also known as the 'Molassis basin'** is made up of soft unconsolidated deposits. Most of the rivers in Nagaland form the tributary of the Brahmaputra. While two rivers of Mizoram and Manipur are the tributaries of the Barak river, which in turn is the tributary of Meghna; the rivers in the eastern part of Manipur are the tributaries of Chindwin, which in turn is a tributary of the Irrawaddy of Myanmar. **Hence statement 2 is not correct.**

- They are flanked by Nepal Himalayas in the west and Bhutan Himalayas in the east. It is relatively small but is a most significant part of the Himalayas. Known for its fast-flowing rivers such as Tista, it is a region of high mountain peaks like Kanchenjunga (Kanchengiri), and deep valleys.

- Q 26. B**
- The best description of the Malthusian Trap is: a cycle where gains in food production are offset by population growth, keeping per capita income at a subsistence level. Proposed by Thomas Malthus, this theory argues that population grows geometrically while food production increases arithmetically, leading to inevitable famine or poverty that keeps living standards low.
 - The Mechanism: Technological improvements or increased food production (e.g., better farming) only lead to temporary increases in income. This temporary prosperity causes population growth to rise.
 - The Trap: Eventually, the higher population consumes the extra food, driving per capita income back down to a subsistence level.
 - The Outcome: The population is kept in check by 'positive checks' such as famine, disease, or war.
 - **Breaking the Trap:** Historically, most of human history was stuck in this trap. It was only broken during the Industrial Revolution in the 19th century, where technological growth finally became so rapid that it outpaced population growth, allowing per capita income to rise consistently for the first time.
 - **Hence, option (b) is the correct answer.**

- Q 27. C**
- **Nilgiri Hills:** Located at the junction of the Western and Eastern Ghats, they are spread across the states of Tamil Nadu, Karnataka, and Kerala. The highest peak is Doddabetta (2,637 m) in Tamil Nadu.
 - **Aravalli Range:** This range stretches across Gujarat, Rajasthan, and Haryana (and extends into the Union Territory of Delhi). It is one of the oldest fold mountain systems in the world. Its highest peak is Guru Shikhar (1,722 m) in Rajasthan.
 - **Rajmahal Hills:** These hills are located entirely within the state of Jharkhand (specifically in the Santhal Pargana division). They are formed from volcanic activity (Rajmahal Traps) and mark the northeastern edge of the Peninsular Plateau.
 - **Cardamom Hills:** They are part of the southern Western Ghats and are situated in Kerala and Tamil Nadu. They form the boundary between southeast Kerala and southwest Tamil Nadu. They are famous for the cultivation of spices like cardamom, pepper, and coffee.
 - **Hence, option (c) is the correct answer.**



Q 28. D The development of pottery in ancient India is a reflection of the cultural and technological developments from the Neolithic period to the later Vedic era.

Neolithic Age:

- **Characteristics:** The earliest pottery was hand-made, unglazed, and often had a rough surface.
- **Materials:** Typically made from clay mixed with mica and sand.
- **Decorative Techniques:** Twisted rice husk cords were sometimes impressed into wet clay for decoration.
- **Examples:** **Coarse grey pottery found at sites like Burzahom.**

Chalcolithic Age:

- **Cultural Distinctions:** Marked by distinct cultures such as Ahar, Malwa, and Jorwe.
- **Pottery Types:**
 - **Black-and-Red Ware:** Commonly used, often with white linear designs.
 - **Black-on-Red Ware:** Characterized by a matte surface.
 - **Ochre Coloured Pottery (OCP):** Identified with the Copper Hoard Culture, found mainly in the upper Ganga Valley.

Harappan Civilization:

- **Advancements:** Pottery became more refined, featuring both polished and unpolished types.

- Techniques: Wheel-thrown pottery became common, often decorated with geometric patterns and motifs.
- Types: Greyware and painted greyware associated with the Vedic people emerged during this period.

Early Vedic Era:

- **Painted Grey Ware (PGW):** This culture emerged during the Rig Vedic period, characterized by painted designs on grey pottery.
- Iron Age Influence: Later Vedic sites show evidence of iron tools alongside PGW.

Later Vedic Era:

- **Northern Black Polished Ware (NBPW):** Marked by glossy, fine pottery used primarily by the affluent as tableware.
- Diversity: The later Vedic people utilized various pottery types including black-and-red ware and painted grey ware.

- Hence option (d) is the correct answer.

- Q 29. C**
- Since 1950, the successive governments at the Centre and in the states have made several laws and formulated various programmes for implementing the Directive Principles, especially socialist principles.
 - **Almost all the states have passed land reform laws to bring changes in the agrarian society and to improve the conditions of the rural masses.** These measures include (a) abolition of intermediaries like zamindars, jagirdars, inamdars, etc; (b) tenancy reforms like security of tenure, fair rents, etc; (c) imposition of ceilings on land holdings; (d) distribution of surplus land among the landless labourers; and (e) cooperative farming.
 - **Various measures have been taken to utilise the financial resources for promoting the common good. These include nationalisation of life insurance (1956), the nationalisation of fourteen leading commercial banks (1969), nationalisation of general insurance (1971), abolition of Privy Purses (1971).**
 - The 44th amendment act did not abolish property completely. 44th Amendment Act of 1978 abolished the right to property as a Fundamental Right by repealing Article 19(1) (f) and Article 31 from Part III. Instead, the Act inserted a new Article 300A in Part XII under the heading 'Right to Property'. It provides that no person shall be deprived of his property except by authority of law. Thus, the right to property still remains a legal right or a constitutional right, though no longer a fundamental right. It is not a part of the basic structure of the Constitution. **Hence, option (b) is the correct answer.**

- Q 30. C**
- Exit polls are voter surveys that any news organization or independent agency undertakes after voting has been concluded to gauge the mood of the people and understand who is likely to emerge as a winner. The exit polls provide information about how votes were cast in the election, the voting pattern and thereby helping in forecasting the winners of the polls.
 - In opinion polls, you can get the information regarding whom the voter plans to vote for while exit polls can help in understanding whom the voter actually voted for in the election. The exit polls can give the electorate as well as the parties some idea about what is to follow post-voting.

- Section 126(1)(b) of R. P. Act, 1951 prohibits displaying any election matter including results of any Opinion Poll or any other poll survey, in any electronic media during the period of 48 hours ending with the hour fixed for conclusion of poll in each of the phases of elections. Hence, statement 1 is correct.
- Section 126A of the R.P. Act 1951, prohibits the conduct of Exit poll and dissemination of their results by means of print or electronic media during the period mentioned therein, i.e. between the hour fixed for the commencement of poll in the first phase and half an hour after the time fixed for close of poll for the last phase in all the States. Hence, statement 2 is correct.

Q 31. B • **Rajya Sabha is a permanent Chamber, not subject to dissolution and one-third of its Members retire every second year. The hallmark of the Rajya Sabha is the principle of continuity as a perpetual House, as a continuous institution in the parliamentary framework.** A scheme of having a Chamber of legislative continuity was needed in a parliamentary system to meet the legislative and constitutional contingency at a time when the popularly elected House may be under dissolution or in the process of reconstitution after the election. Hence, option (b) is the correct answer.

- As the Rajya Sabha is a permanent House, the dissolution of Lok Sabha has limited impact on the Bills pending before the Rajya Sabha. A Bill which is pending in the Rajya Sabha and has not been passed by the Lok Sabha, will not lapse on the dissolution of the Lok Sabha. Thus, the continuity of the Rajya Sabha ensures a significant measure of legislative continuity.

Q 32. C • **Just when the HSRA revolutionaries had begun to move away from individual heroic action, the death of Sher-i-Punjab Lala Lajpat Rai due to lathi blows received during a lathi-charge on an anti-Simon Commission procession (October 1928) led them once again to take to individual assassination. Bhagat Singh, Azad and Rajguru shot dead Saunders, the police official responsible for the lathicharge in Lahore. Hence, option (c) is the correct answer.**

- The assassination was justified with these words: “The murder of a leader respected by millions of people at the unworthy hands of an ordinary police officer...was an insult to the nation. It was the bounden duty of young men of India to efface it... We regret to have had to kill a person but he was part and parcel of that inhuman and unjust order which has to be destroyed.
- Bhagat Singh and Batukeshwar Dutt threw a bomb in the Central Legislative Assembly on April 8, 1929 to protest against the passage of the Public Safety Bill and Trade Disputes Bill aimed at curtailing civil liberties of citizens in general and workers in particular. The bombs had been deliberately made harmless and were aimed at making ‘the deaf hear’. The objective was to get arrested and to use the trial court as a forum for propaganda so that people would become familiar with their movement and ideology.

Q 33. A • **Major Famine Commissions and Their Roles:**

- **Campbell Commission (1866):** Formed after the Odisha famine, it examined the causes and, notably, blamed government inaction for the high mortality.
- **Strachey Commission (1878–1880):** Headed by Sir Richard Strachey under Lord Lytton, it was the first major commission. It recommended state intervention in the food trade and led to the creation of the Famine Code in 1883. **Hence, point 1 is correct.**

- Lyall Commission (1888): Appointed to review the effectiveness of existing famine policies. **Hence, point 4 is correct.**
- **MacDonnell Commission (1900):** Led by Sir Anthony MacDonnell, it examined the 1896-97 and 1899-1900 famines, focusing on improving relief administration and criticizing the "moral strategy" of offering minimal relief. **Hence, point 2 is correct.**
- Famine Inquiry Commission / Woodhead Commission (1944): Established to investigate the 1943 Bengal famine. It highlighted the failures in food distribution and policy, although it was criticised for understating the colonial government's responsibility.
- **Lee Commission:** The Lee Commission was appointed in 1923 to look into the organisation and general conditions of service as well as the methods of recruitment for Europeans and Indians in the civil services. Being concerned only with the superior civil services, it came to be known as the Royal Commission on the Superior Civil Services in India. **Hence, point 3 is not correct.**
- **Hence, option (a) is the correct answer.**

- Q 34. B**
- The Rashtrakuta Dynasty (c. 753–982 CE) was a powerful Indian imperial dynasty that dominated the Deccan and large parts of the Indian subcontinent for nearly two centuries.
 - The Rashtrakuta Kingdom was established by Dantidurga (also known as Dantivarman). In the mid-8th century, Dantidurga overthrew the Chalukyas of Badami (specifically Kirtivarman II) to lay the foundation of the dynasty. Krishna III was actually the last great ruler of the dynasty, reigning much later in the 10th century. **Hence, statement 1 is not correct.**
 - Manyakheta (modern-day Malkhed in Karnataka) served as the imperial capital of the Rashtrakuta Empire. It was established as the capital during the reign of Amoghavarsha I. **Hence, statement 2 is correct.**
 - The magnificent rock-cut Kailasa temple at Ellora (Cave 16) was built during the Rashtrakuta rule. Specifically, it was commissioned by King Krishna I in the 8th century. **Hence, statement 3 is correct.**

- Q 35. C**
- To address the difficulties faced by the working mothers in giving due child care and protection to their children, **day-care crèche facilities** are being provided through Palna Scheme.
 - In 2022, erstwhile National Creche Scheme was reorganized and renamed as Palna Scheme under the sub scheme 'Samārthya' of 'Mission Shakti'.
 - Palna is a **Centrally Sponsored Scheme** ensuring the participation of State/ UT government to ensure better day-to-day monitoring and proper implementation of scheme, and is implemented with a funding ratio of 60:40 between Centre and State Governments and UTs with legislature except North East & Special Category States where ratio is 90:10. For UTs without legislature, 100% funding is provided by the central government.
 - **The objective of the Palna scheme is to provide quality crèche facility in safe and secure environment for children (from ages 6 months – 6 years), nutritional support, health and cognitive development of children, growth monitoring & immunization. Crèche facilities under Palna are provided to all mothers, irrespective of their employment status.**
 - **Apart from the aforementioned primary objective, another objective of the component would involve monitoring compliance of the stipulations laid down in Section 11A of the Maternity Benefit Act regarding setting up of crèche facilities by the establishments.** For this purpose, the States/UTs would explore the possibility of hosting an application/portal in convergence with the

Labour & Employment Departments of their respective States/UTs for facilitating establishments to register on the portal and furnish details in respect of the crèches set up in accordance with the Act. As part of enhancing access to childcare facilities, the Ministry of Labour & Employment has amended the Maternity Benefit Act to mandate all establishments with 50 or more employees to provide a crèche facility.

- **Crèche services formalise the child care responsibilities hitherto considered as part of domestic work. Formalization of care work supports the “decent work campaign” to achieve Sustainable Development Goal 8 – Decent work and economic growth. This will also enable more mothers, who will be free from unpaid childcare responsibilities, to take up gainful employment.**
- **Hence, option (c) is the correct answer.**



- Q 36. B**
- The term "Appropriate Government" refers to the government entity—central or state—that has the authority or jurisdiction to make decisions, enact laws, or administer policies in specific contexts as defined by the constitution or legislation. It is a term often used in legal texts to specify which level of government is responsible for a given area of governance, taking into account the division of powers between different levels of government.
 - In legal terms, particularly under Indian law, "appropriate government" typically denotes the Central Government or the State Government, depending on the jurisdiction or nature of the establishment involved.
 - **Hence option (b) is the correct answer.**

- Q 37. D**
- In early 1937, elections to provincial assemblies were announced and once again the debate on the future strategy to be adopted by the nationalists began. Everyone in the Congress agreed that the 1935 Act was to be opposed root and branch but it was not clear how it was to be done in a period when a mass movement was not yet possible. **Jawaharlal Nehru, Subhash Bose, and Congress socialists and communists were opposed to office acceptance** and thereby in the working of the 1935 Act because they argued that it would negate the rejection of the Act by the nationalists. As a counter-strategy, the **leftists proposed entry into the councils with an aim to create deadlocks**, thus making the working of the Act impossible (older Swarajist strategy). And, as a long-term strategy, they advocated an increased reliance on workers and peasants, integration of their class

organisations into the Congress, thus imparting a socialist direction to the Congress and preparing for the resumption of a mass movement. **Hence statement 1 is correct.**

- Gandhi opposed office acceptance in the CWC meetings but by the beginning of 1936, he was willing to give a trial to the formation of Congress ministries. In its sessions at Lucknow in early 1936 and **Faizpur in late 1937**, the Congress decided to fight elections and postpone the decision on office acceptance to the postelection phase. **Hence statement 2 is correct.**
- The Congress did extremely well in the 1937 elections, winning 711 out of 1585 provincial assembly seats, with **absolute majorities in five provinces out of eleven** (Madras, Bihar, Orissa, C.P. and U.P.) and a **near-majority in Bombay** (86 out of 175). Official backing failed to save from utter rout loyalist landlord-based groups like the Nawab of Chhatari's **National Agriculturist Party** in U.P. and the **Justice Party** in Madras. Even the **poor showing in the Muslim constituencies** (the Congress contested only 58 out of 482 reserved seats, and won 26) was somewhat counterbalanced by the evident failure of the Muslim League to make good its claim to be the sole representative of the Muslims. The **League failed to win a single seat in the N.W.F.P. and could capture only 2 out of 84 reserved constituencies in the Punjab and 3 out of 33 in Sind**. The Congress also won most scheduled caste seats, except in Bombay where **Ambedkar's Independent Labour Party** captured 13 out of 15 seats reserved for Harijans. **Hence statement 3 is correct.**
- **Hence option (d) is the correct option.**

- Q 38. B**
- **Kundhei, Orissa:** The string puppets of Orissa are known as Kundhei. Made of light wood, the Orissa puppets have no legs but wear long flowing skirts. They have more joints and are, therefore, more versatile, articulate and easy to manipulate. The puppeteers often hold a wooden prop, triangular in shape, to which strings are attached for manipulation. The costumes of Kundhei resemble those worn by actors of the Jatra traditional theatre. The music is drawn from the popular tunes of the region and is sometimes influenced by the music of Odissi dance.
 - **Putul Nautch, West Bengal:** The traditional rod puppet form of West Bengal is known as Putul Nautch. They are carved from wood and follow the various artistic styles of a particular region. In Nadia district of West Bengal, rod-puppets used to be of human size like the Bunraku puppets of Japan. This form is now almost extinct. The Bengal rod-puppets, which survive are about 3 to 4 feet in height and are costumed like the actors of Jatra, a traditional theatre form prevalent in the State. These puppets have mostly three joints. The heads, supported by the main rod, is joined at the neck and both hands attached to rods are joined at the shoulders.
 - **Yampuri, Bihar:** The traditional Rod puppet of Bihar is known as Yampuri. These puppets are made of wood. Unlike the traditional Rod puppets of West Bengal and Orissa, these puppets are in one piece and have no joints. As these puppets have no joints, the manipulation is different from other Rod puppets and requires greater dexterity.

- Q 39. D**
- Sufis were divided into different **silsilahs (orders)** with each silsilah having its **own pir (guide)** called **Khwaja or Sheikh**. **Hence pair 1 is correctly matched.**
 - The pir and his disciples lived in a **khanqah (hospice)**. **Hence pair 2 is correctly matched.**
 - A pir nominated a successor or wali from his disciples to carry on his work.
 - The Sufis organized **samas (a recital of holy songs)** to arouse mystical ecstasy. **Hence pair 3 is correctly matched.**

- Q 40. C** • **The Central Vigilance Commission** is the apex integrity institution of the Govt. of India, mandated to combat corruption and ensure integrity in administration and public services. **The CVC is not controlled by any Ministry/Department. It is an independent body which is only responsible for the Parliament. The mandate of the Commission is laid down in the CVC Act, 2003. Some of the roles and functions of CVC:**
- The Commission also conducts preliminary inquiry into the complaints referred by Lokpal in respect of Gr. A, B, C& D officials.
 - The appointment of the officers to the posts of the level of Deputy Director and above (including the Director) in the Directorate of Enforcement is recommended by a committee under the chairpersonship of Central Vigilance Commissioner and with Vigilance Commissioner and other officers of Government of India as members. **Hence, option (c) is the correct answer.**
- **National Commission for Minorities:** The enactment of the National Commission for Minorities Act, 1992, the Minorities Commission became a statutory body and renamed as National Commission for Minorities. **The National Commission for Minorities (NCM) in India comes under the Ministry of Minority Affairs.**
- **The Central Information Commission has been constituted with effect from 12-10-2005 under the Right to Information Act, 2005.** The jurisdiction of the Commission extends over all Central Public Authorities. This Commission consists of the Chief Information Commissioner and Information Commissioners not exceeding 10. **It comes under the Ministry of Personnel, Public Grievances and Pensions.**
- **The National Disaster Management Authority (NDMA),** headed by the Prime Minister of India, is the apex body for Disaster Management in India. **It comes under the Ministry of Home Affairs (MHA), Government of India.**
- Q 41. D** • The 14th edition of the India - Thailand joint military Exercise MAITREE-XIV commenced with a grand opening ceremony at Joint Training Node (JTN), Umroi, Meghalaya. The exercise was conducted from 1st to 14th September 2025. The bilateral exercise is part of the ongoing military-to-military exchange programme between the two countries and aims to enhance cooperation, interoperability, and mutual understanding between the Indian Army and Royal Thai Army. The 13th edition of the exercise was held at Fort Vachiraprakan in Tak Province, Thailand. **Hence, statement 1 and 2 are correct.**
- **The joint exercise will focus on company level counter-terrorist operations in semi-urban terrain under Chapter VII of the United Nations Charter. Hence, statement 3 is correct.**
- Exercise MAITREE, instituted in 2006, is one of the important joint training exercises between India and Thailand.
- Q 42. B** • **Saranjami System under Marathas was similar to the Jagir system of Mughals where lands were granted (non-hereditary) for maintenance of troops.** The land was mostly in the form of a rural Watan (rights given in reward for previous service or merit) or Jagir, its owner being entitled to extract revenue from the villages included in the territory. Saranjamdar was the title given to the landlord or holder of a Saranjam. Usually it was bestowed on that person for heroic deeds in the military field, thus, most Saranjamdars were former military officers. **Hence option (b) is the correct answer.**

- Q 43. C** • The Turkish state in India was militaristic and aristocratic. The Turkish nobles tried, at first, to monopolise the high offices of state denying a share to the Tajiks, Afghans, and other non- Turkish immigrants. The nobility acquired a broader base under the Tughlaqs. However, a noble birth still remained a very important qualification. The vast majority of the Muslims as well as the Hindus had, therefore, little opportunity for occupying high offices of state. The Hindus dominated trade and constituted the ruler aristocracy, and the lower administrative wing without whose cooperation the state could not function. A kind of tacit sharing of power between the rural hindu aristocracy, and the city-based administrators was, thus, a factor of capital importance for the Delhi Sultanate.
- As for the Hindu subjects, from time of the Arab invasion of Sindh, they had been given the status of **zimmis or protected people, i.e. those who accepted the Muslim rule and agreed, to pay tax called jizyah**. This was a tax in lieu of military service, and was paid on a graduated scale according to means, women, children and the indigent who had insufficient means being exempt from it. The brahmanas also remained exempt, though this was not provided for in the Muslim law. At first jizyah, was collected along with the land revenue. In fact it was difficult to distinguish jizyah from land revenue since cultivators were Hindus. **Later Firuz, while abolishing many illegal cesses, made jizyah a separate tax. He levied it on the Brahmana also. Hence, option (c) is the correct answer.**

- Q 44. A** • The Intergovernmental Negotiations framework or IGN is a group of nation-states working within the United Nations to further reform of the United Nations Security Council (UNSC).
- The IGN is composed of several different international organizations, including the African Union; the G4 nations; the Uniting for Consensus Group (UfC), also known as the "Coffee Club"; the L.69 Group of Developing Countries; the Arab League and the Caribbean Community (CARICOM).
- **Uniting for Consensus is a movement, nicknamed the Coffee Club, that developed in the 1990s in opposition to the possible expansion of permanent seats in the United Nations Security Council.**
- **Hence option (a) is the correct answer.**

- Q 45. A** Secondary Particulate Matter (SPM) is not emitted directly from sources like exhaust pipes or chimneys. Instead, it forms in the atmosphere through chemical reactions between gaseous precursors.
- Sulphur dioxide: Oxidizes in the atmosphere to form sulphuric acid, which then reacts with other compounds (like ammonia) to form sulphate particles. **Hence, point 1 is correct.**
- Nitrogen oxides: React with moisture and other chemicals to form nitric acid, leading to the creation of nitrate particles. **Hence, point 2 is correct.**
- Ammonia: Acts as a crucial neutralizing agent. It reacts with sulphuric and nitric acids to form ammonium sulphate and ammonium nitrate, which are major components of fine particulate matter. **Hence, point 3 is correct.**

While CO is a major primary pollutant and can participate in atmospheric chemistry (such as the formation of ground-level ozone), it is **not a direct precursor to the formation of secondary particulate matter**. It remains in a gaseous state and does not condense or react into solid/liquid particles under typical atmospheric conditions. **Hence, point 4 is not correct.**

- Q 46. C** • **Milk Production and Export:** While India has led global milk production since the late 1990s, countries like the European Union (collectively) or New Zealand are the top exporters. High

domestic demand and strict international quality standards (SPS) often limit India's dairy exports.

Hence option (c) is the correct answer.

- **Rice:** India is the world's largest exporter of rice, including both Basmati and non-Basmati varieties, often commanding over 40% of the global market share.
- **Turmeric:** India dominates this market as the largest producer, consumer, and exporter, holding roughly 80% of the global trade share.
- **Cumin:** India is the leading global exporter of cumin, producing nearly 70% of the world's supply

Q 47. A Background — the Second Round Table Conference (1931):

- The Second Round Table Conference (7 September – 1 December 1931, London) followed the Gandhi–Irwin Pact and was the only one of the three conferences that included the Indian National Congress. Mahatma Gandhi attended as the Congress's sole official representative; the conference was intended to discuss constitutional reforms and communal questions, but it failed to reach a settlement acceptable to the major Indian parties.

Who represented the Government of India?

- The Government of India (i.e. the British Indian administration) sent an official delegation to the conference. Contemporary lists of delegates name **C. P. Ramaswami Iyer, Narendranath Law, and M. Ramachandra Rao among the Government of India representatives. Hence, option (a) is the correct answer.**
- Some official rosters and later summaries also include names such as Bhupendra Nath Mitra in the broader British-Indian team, but the three names in option (a) are accurately recorded as Government of India delegates for the Second RTC.

Other participants

- The Conference brought together a wide array of Indian interests: Congress (Gandhi), princely states, communal delegations (Muslims, Sikhs, Anglo-Indians), business and professional groups, and official British-Indian delegates. The presence of official Government of India delegates — rather than only metropolitan British politicians — reflected the technical and administrative nature of many agenda items (federation, defence, finance), and explained why the Government of India delegation was kept on record in the proceedings.

Q 48. B • Larsen & Toubro (L&T) has won the deal from India's Department of Atomic Energy (DAE) to build a LIGO at Aundha in Hingoli district, Maharashtra.

- The LIGO-India project is being built in the Hingoli district of Maharashtra (specifically near Aundha Nagnath), not in the Theni district of Tamil Nadu. The project located at Pottipuram in Theni, Tamil Nadu, is the India-based Neutrino Observatory (INO). **Hence, statement 1 is not correct.**
- LIGO (Laser Interferometer Gravitational-Wave Observatory) uses the physical properties of light (laser interferometry) and space-time (detecting minute distortions or "ripples") to observe and understand the origins of gravitational waves. **Hence, statement 2 is correct.**
- LIGO-India is India's first dedicated Gravitational Wave Observatory. It will be part of a global network and is expected to be operational around 2030. **Hence, statement 3 is correct.**

Principle: LIGO exploits the physical properties of light and of space itself to detect and understand the origins of gravitational waves.

- LIGO detects gravitational waves by using laser interferometry to measure minuscule changes in space caused by passing waves.
- It uses twin 4-km-long L-shaped vacuum tunnels to split a laser beam, reflect it off mirrors, and recombine them, creating an interference pattern that reveals stretching or compressing of space.

Q 49. B Dhands of Rajasthan: The Saline Playas of the Thar Desert

- In the diverse physical geography of India, regional landforms often carry unique local names that reflect centuries of human-environment interaction. One such term, “dhands,” holds specific significance in the arid landscape of western Rajasthan.

What Are Dhands?

- **The term ‘dhand’ refers to shallow saline depressions or playa lakes found primarily in the Thar Desert region.** These geographical features are essentially seasonal wetlands or desert lakes that occupy saucer-shaped, low-lying basins, which fill up with water during the monsoon season. **Hence, option (b) is the correct answer.**
- Over time, due to high evaporation rates and minimal outflow, these lakes become saline and support salt extraction industries. During dry seasons, they often shrink or dry up entirely, leaving behind expansive salt-encrusted surfaces.

Aeolian Origins

- The formation of dhands is typically attributed to Aeolian processes—shaped by wind action in sandy desert environments. Blowing winds over long periods erode the loose sand, forming shallow depressions which later accumulate rainwater and dissolved minerals, especially salts.

Notable Dhands in Rajasthan

- Some of the best-known examples of dhands in India include:
 - • Sambhar Lake – the largest inland salt lake in India
 - • Didwana, Kuchaman, Pachpadra, and Phalodi lakes – all important for regional salt production and biodiversity during wet phases
- These dhands serve as temporary habitats for migratory birds and hold ecological, economic, and cultural value in the desert economy.

What Dhands Are Not

- It’s important to distinguish dhands from other landforms:
 - • They are not tidal channels or estuarine connectors (commonly known as creeks).
 - • They are not oxbow lakes, which are typically formed by meandering rivers.
 - • Nor are they glacial lakes, which occur in high Himalayan altitudes due to melting ice.

- Q 50. A**
- A judge in the Calcutta High court recently joined a political party. This has created a debate over judicial independence and free and fair working.
 - This is not the first time for a judge to become a member of political party or contest elections- from Lok Sabha to Presidential elections. Mohammad Hidayatullah was the Chief Justice of India from 1968 to December 16, 1970. He served as the vice president of India for a brief while in 1969 when President Dr Zakir Hussain died. In 1978, he was elected vice president unopposed and served from August 1979 to August 1984. MC Chagla, a legendary judge of the Bombay High Court, formed his

own party, the Muslim Nationalist Party. In 1948, he became Chief Justice of the Bombay High Court and remained in that position till 1958. After his stint as a judge, he served as India's education minister from 1963 to 1966 and the country's external affairs minister till September 1967. The Constitution or any law does not bar such actions by retired judges.

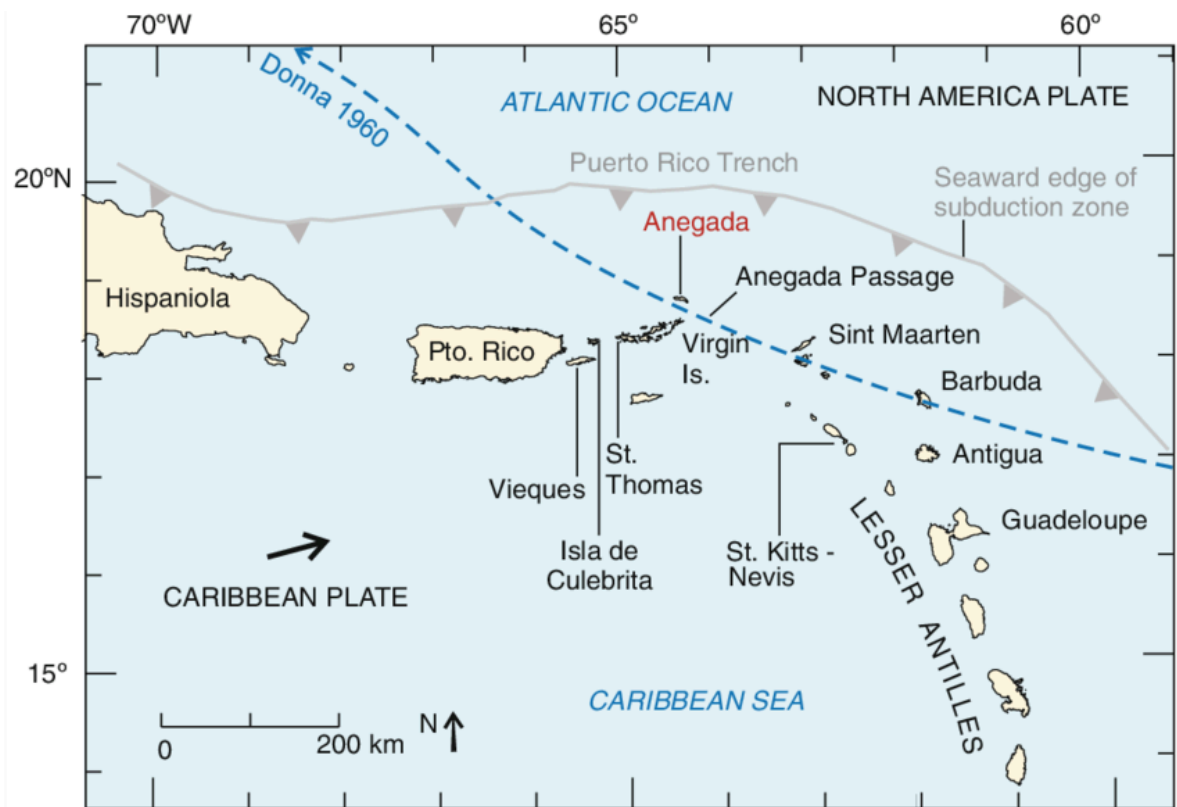
Technically speaking, the Constitution does not bar a person with a political affiliation or a person who had been a politician from becoming a judge. For the high courts, Article 217 (2) of the Constitution of India states: "A person shall not be qualified for appointment as a Judge of a High Court unless he is a citizen of India and —

- **has for at least 10 years held a judicial office in the territory of India; or**
- **has for at least 10 years been an advocate of a High Court or of two or more such courts in succession." Hence, statement 1 is correct.**
- These are just broad parameters; the task of selecting candidates to be made judges in the Supreme Court and the high courts is left largely to the collegium. The memorandum of procedure (MoP), a playbook for the appointment of judges between the government and the judiciary, prescribes the mechanism and timeline for the appointment of judges to high courts.
- **While the Constitution and the MoP speak of eligibility and procedure, it does not mention that members having political affiliations should not be made judges. Thus, it is clear that there is no legal bar on people with political links to becoming judges. However, the collegium and law minister can object to a person's recommendation owing to their political affiliations. Hence, statement 2 is not correct.**

Q 51. A Context: Fighter jets and guided-missile destroyers transited the Anegada Passage as a part of USA's Operation Southern Spear against Narco Terrorism in the Caribbean.

Anegada Passage

- The Anegada Passage (also known as the Anegada Trough) is a critical 40-mile-wide maritime channel that serves as a primary gateway between the Atlantic Ocean and the Caribbean Sea.
- It separates the British Virgin Islands (west) from the Leeward Islands (southeast).
- It has the greatest depth (more than 7,550 feet [2,300 m]) of any channel in the eastern Caribbean. The passage is one of the two through which subsurface water enters the Caribbean (the other being the Windward Passage).



Hence, option (a) is the correct answer.

Q 52. B Namo Bharat RRTS

- It is a new rail-based, semi-high-speed (up to 180 km/ hr), commuter transit system
- It is different from Metro as it caters to longer-distance travel with fewer stops at higher speed.
- It is different from conventional trains as it provides high-frequency, point-to-point regional travel on a dedicated pathway.
- Implementing agency: National Capital Region Transport Corporation (joint venture between the centre and Rajasthan, UP, Haryana, Delhi).

Significance of RRTS

- Addressing urban Congestion: More than 40% urban population to live in urban areas by 2030.
- Reducing travel Time and enhanced labour productivity e.g., Delhi Merrut RRTS to reduce travel time to 60 minutes from 3 hours.
- Balanced Regional development -By encouraging commuters to live in suburban and regional areas it supports polycentric economic development.
- Environmental Sustainability: apart from being electrically-powered, it can reduce reliance on private cars.
- Social Empowerment: The project is a landmark for "Nari-Shakti," with women comprising the majority of train operators and station control staff.

Hence, option (b) is the correct answer.

Q 53. B • The Rajasthan Plains are part of the Thar Desert and its semi-arid margins.

- Vegetation: Because the annual rainfall is generally less than 50 cm, the predominant vegetation is Tropical Thorn Forest. These plants are xerophytic, meaning they have adapted to conserve water through thorns, thick waxy leaves, and long taproots (e.g., Khejri, Ber, Acacia). **Hence, pair 1 is correctly matched.**
- The Andaman & Nicobar Islands are located in the equatorial belt. They receive heavy rainfall (exceeding 250 cm) and high temperatures year-round. This leads to the growth of dense Tropical Evergreen Forests. These forests are characterized by a multi-layered structure with a thick canopy that prevents sunlight from reaching the ground. **Hence, pair 2 is correctly matched.**
- The Meghalaya Plateau (including the Garo, Khasi, and Jaintia hills) is famous for being one of the wettest regions in the world. Due to the extremely high rainfall, the predominant vegetation is Tropical Evergreen and Semi-Evergreen Forest.
- Temperate refers to higher latitudes or much higher altitudes (like the Himalayas above 2500m). Dry Deciduous is found in regions with much lower rainfall (70-100 cm). **Hence, pair 3 is not correctly matched.**

Q 54. D According to Article 167, it shall be the duty of the Chief Minister:

- to communicate to the governor of the state all decisions of the council of ministers relating to the administration of the affairs of the state and proposals for legislation;
- to furnish such information relating to the administration of the affairs of the state and proposals for legislation as the governor may call for; and
- if the governor so requires, to submit for the consideration of the council of ministers any matter on which a decision has been taken by a minister but which has not been considered by the council. **Hence, option (d) is the correct answer.**

- Q 55. B**
- The Great Blue Hole is a giant marine sinkhole located off the coast of Belize in the Caribbean Sea, near the Lighthouse Reef Atoll.
 - It was formed when limestone caves collapsed during periods of lower sea levels in the last Ice Age and were later submerged as sea levels rose.
- Key features:
- Nearly 300 m wide and ~125 m deep
 - Formed due to karst processes
 - Recognised as a UNESCO-associated marine feature (part of Belize Barrier Reef System).
- **Hence, option (b) is the correct answer.**
 -



Q 56. D • Investor Education and Protection Fund or IEPF is a fund set up under Section 205C of the Companies Act, 1956 by the Ministry of Corporate Affairs in 2016 to pool all the dividends of the Asset Management Companies, matured deposits, share application interests or money, debentures, interests, etc. that are unclaimed for seven years.

- All the money collected from the mentioned sources has to be transferred to IEPF. Investors, who are trying to seek a refund for their unclaimed rewards can do so from the Investor Protection and Education Fund (IEPF).
- **Thus the source of funds is through unclaimed shares, dividends, matured deposits/debentures etc, and not budgetary allocations. Hence, statement 1 is not correct.**
- **Investor Education And Protection Fund Authority is entrusted with the responsibility of administration of the Investor Education Protection Fund (IEPF). Hence, statement 2 is not correct.**

Objectives of Investor Education and Protection Fund (IEPF)

- Educating investors about how the market operates.
- Making investors educated enough so that they can analyze and make informed decisions.
- Educating investors about the volatility of the markets.
- Making investors realize their rights and various laws about Investing.
- Promoting research and surveys to spread knowledge among the investors.

Q 57. A • Certification marks on a commercial product are usually called a mark of validity or an assurance of the fact that the manufacturer has tested the product and it has ensured that it meets the given quality standard.

- It also helps to specify the location of the origin, materials of construction, mode of production, characteristics of the good etc.
- Some of these marks are mandatory while some are advisory.
- A certification mark certifies the nature or origin of the goods or the services to which it has been applied.
- **Bureau of Indian Standards (BIS)**, which comes under the aegis of the International Organisation for Standardisation(ISO standards), govern all the industrial standardization and industrial product certifications in India. It is thus a national standards organization of India.

BIS Hallmark:

- The BIS hallmark is a hallmarking system for the sale of precious metal like gold and silver jewellery in India which certify the purity of the metal.
- It verifies that the jewellery conforms to the standards which are set by the Bureau of Indian Standards.

ECO Mark:

- The Government of India has instituted a scheme for labelling environment-friendly products to be known as ECO Mark. **The scheme is being administered by the Bureau of Indian Standards.** The Eco mark Scheme covers various product categories like Soaps and Detergents, Paints, Food Items, lubricating oils, Packaging materials/Package, Architectural Paints and Powder Coatings, Batteries, Electrical and electronic goods, Food Additives, Wood Substitutes, Cosmetics, Aerosols and Propellants, Plastic Products, Textiles, Fire-extinguisher, Leather and Coir & Coir Products
- The Scheme is being operated on a national basis and provides certification and labelling for household and other consumer products which meet certain environmental criteria along with quality requirements prescribed in relevant Indian Standards for the product.

FPO Mark:

- In India, all processed and packaged fruit products are mandated to be certified with FPO Mark. Food process order or **FPO mark is issued by the Ministry of Food Processing Industry** which also develops the standards for this mark. The mark was named after the law called Fruit Products offer. FPO certification in India is effective since 1955 but became mandatory only in 2006 after the Food Safety and Standards Act. **Hence option (a) is the correct answer.**

- Q 58. B**
- Robert Clive introduced the Dual Government system in Bengal in 1765. Under this arrangement, the administration was divided between the East India Company and the Nawab of Bengal.
 - Battle of Plassey (June 23, 1757): Robert Clive's victory over Siraj-ud-Daulah laid the territorial foundation of British rule in India. **Hence, statement 2 is correct.**
 - Robert Clive was the first Governor of Bengal, but he was never the Governor-General. The office of the Governor-General of Bengal was created later by the Regulating Act of 1773, and Warren Hastings was the first to hold this title. **Hence, statement 1 is not correct.**
 - Robert Clive returned to India in 1765 after the British victory at the Battle of Buxar, which solidified British dominance. In August 1765, Robert Clive brokered two key treaties at Allahabad (Treaty of Allahabad), one with Nawab Shuja-ud-Daula of Awadh and the other with Mughal Emperor Shah Alam II.
 - Shuja-ud-Daula ceded territories and paid a war indemnity of Rs 50 lakh, while Shah Alam II granted the EIC the Diwani of Bengal, Bihar, and Orissa in exchange for an annual payment of Rs 26 lakh and Rs 53 lakh for Nizamat functions (military defence, police duties, and the administration of justice). **Hence, statement 3 is correct.**

- Q 59. B**
- The Sukinda Valley in Odisha, India, contains approximately 97–98% of the country's chromite deposits, making it one of the largest opencast chromite ore mining areas in the world. Located in the Jajpur district, this 25 km long valley is the primary source of chromium for India's stainless steel and refractory industries, though mining activities have caused severe, widespread hexavalent chromium contamination in the water, air, and soil.

- Hence, option (b) is the correct answer.



- Q 60. A**
- **Statement 1 is incorrect.** Stagflation is defined as a combination of stagnant economic growth (or recession and not expansion), high inflation, and high unemployment, not an expanding economy without employment. It is a period of falling, not expanding, economic activity.
 - **Statement 2 is correct.** The Phillips Curve originally suggested an inverse relationship between unemployment and inflation (or economic growth/expansion and unemployment). Stagflation—the combination of high inflation and high unemployment—directly contradicts the traditional Phillips Curve, which implies that high inflation should only occur when unemployment is low and the economy is rapidly growing.

- Q 61. C** The government has given a go-ahead to the production of next generation airborne early warning and control system (AWACS) India programme also referred to as the Netra MkII.

Netra MkII

- It will be mounted on the Airbus A321 platform.
- Key Features: Detect, track, and monitor threats across vast distances.
- Much more than an aircraft as it is a flying command centre which will extend our national vigilance.

Hence, option (c) is the correct answer.

- Q 62. A** • **Recent Context:** Recently, a 2-Judge bench of the Supreme Court held that bar on anticipatory bail under Section 18 of the SC and ST (Prevention of Atrocities) Act, 1989 is not applicable unless a prima facie case under the Act is made out against the accused. Section 18 of the Act states that Section 438 of the CrPC, which provides for anticipatory bail, shall not apply in relation to cases involving offence under the Act.

Anticipatory bail:

- It is a direction by High Court or the Court of Session to release a person on bail, apprehending arrest on accusation of having committed a non-bailable offence. Hence, option (a) is the correct answer.
- Section 438 of the Code of Criminal Procedure (CrPC) contained provisions related to anticipatory bail.
- Under reformed criminal laws, Section 482 of Bhartiya Nagarik Suraksha Sanhita (2023) contains relevant provisions for anticipatory bail.

- Q 63. B** • The Equator indeed receives intense solar radiation, which causes high evaporation, it also experiences heavy daily rainfall (convective rain) and high humidity. The fresh rainwater dilutes the surface seawater, keeping the salinity relatively lower.
- The highest surface salinity is actually found in the sub-tropical regions (20°–30° North and South), where evaporation is extremely high due to clear skies, but rainfall is minimal. **Hence, statement 1 is not correct.**
 - Generally, salinity decreases as we move toward the poles because of low evaporation and the melting of ice (freshwater influx). However, the North Sea is an exception as the North Atlantic Drift (an extension of the warm Gulf Stream) carries saltier, warmer water from the tropical Atlantic towards the higher latitudes of Europe. This saline influx keeps the North Sea's salinity higher than other regions at similar latitudes. **Hence, statement 2 is correct.**



- Q 64. B • Therukoothu is the most popular form of folk drama of **Tamil Nadu**, literally means "street play". It is mostly performed at the time of annual temple festivals of Mariamman (Rain goddess) to achieve a rich harvest. **Hence statement 1 is not correct.**
- At the core of the extensive repertoire of Therukoothu, there is a **cycle of eight plays based on the life of Draupadi**. **Hence statement 2 is correct.**
 - **Kattiakaran**, the Sutradhara of the Therukoothu performance, gives the gist of the play to the audience and **Komali** entertains the audience with his buffoonery.
 - The antiquity of Therukoothu is unknown. Being a generic term in Tamil for play, 11 types of koothu are mentioned in **Silappadikkaram**, but Therukoothu has remained the only total theatre with the **actor having to sing, dance, speak and perform rituals**.
- Q 65. D • Fiat money is a government-issued currency that is **not backed by a physical commodity, such as gold or silver**, but rather by the government that issued it.
- It gets its **value based on the trust people place in the authorities that issue it**. Commodity-backed currencies, on the other hand, get their value from the underlying price of the gold, silver, or other materials they're linked to.
 - **It has no intrinsic value like gold or silver.**
 - **It is called legal tender as it cannot be refused by any citizen of the country for settlement of any kind of transaction.**
 - **Hence option (d) is the correct answer .**
- Q 66. B • Article 266 of the Constitution defines the Public Account as being those funds that are received on behalf of the Government of India. Public Account funds do not belong to the government and have to be finally paid back to the persons and authorities that deposited them. Parliamentary authorisation for such payments is not required. However, when money is withdrawn from the

Consolidated Fund with the approval of Parliament and kept in the Public Account for expenditure for a specific purpose, it is submitted for a vote in Parliament.

- There are five major heads of accounts under the Public Account: (i) **Small Savings, Provident Fund and Other Accounts** (ii) **Reserve Funds** (iii) **Deposits and Advances** (iv) **Suspense and Miscellaneous** and (v) **Remittances**. (**Treasury bills are the part of Consolidated fund of India**).
- The first three of these accounts deal with receipts and payments in respect of which the Government is liable to repay the moneys received or has a claim to recover the amounts paid. In respect of these transactions, the Government acts as a banker, receiving amounts which it later repays and paying out advances which it subsequently recovers. Provident Funds of Government Employees, Deposits of Local Funds, **Reserve Funds Deposits made by outside agencies**, Departmental Advances, etc. fall under this category. The other two accounts—Suspense and Remittances - are used only for adjustment purposes.

Q 67. D UNCLOS Maritime Zones

- **Territorial Sea:** Up to 12 nm from baseline; full sovereignty applies.
- **Contiguous Zone:** 12–24 nm; limited control for specific laws (customs, immigration, etc.).
- **Exclusive Economic Zone (EEZ):** Up to 200 nm; exclusive rights for resource exploration and exploitation, not full sovereignty.
- **Analysis of Statements:**
- **Statement (I):** Incorrect - Incorrect. Territorial Sea extends up to 12 nm; 24 nm is the outer limit of the Contiguous Zone.
- **Statement (II):** Incorrect - Incorrect. EEZ confers exclusive economic rights, not full sovereignty like land territory.

Hence, the correct answer is **(d) Neither I nor II**.

- Q 68. A**
- There has been a rising concern about the impact of the Digital Data Protection Act 2023 on the Right to information of the people. The DPDP Act received the President's assent on August 11, 2023, and will come into force after the Rules under the Act are notified.
 - The DPDP Act is intended to "provide for the processing of digital personal data in a manner that recognises both the right of individuals to protect their personal data and the need to process such personal data for lawful purposes and for matters connected therewith or incidental thereto".
 - Section 44(3) of the DPDP Act mentions a change that will be made in Section 8(1)(j) of the RTI Act. This change, RTI activists say, will greatly reduce the amount of information that government agencies are obligated to disclose under the 2005 Act.
 - **Section 8(1)(j) of RTI Act states: "...There shall be no obligation to give any citizen... information which relates to personal information the disclosure of which has no relationship to any public activity or interest, or which would cause unwarranted invasion of the privacy of the individual unless the Central Public Information Officer or the State Public Information Officer or the appellate authority...is satisfied that the larger public interest justifies the disclosure of such information."**
 - **Section 44(3) of the DPDP Act shortens this clause in the RTI Act to broaden the scope for denying information. It says: "In section 8 of the Right to Information Act, 2005, in sub-**

section (1), for clause (j), the following clause shall be substituted, namely: — “(j) information which relates to personal information”.

- Hence, there is a potential impact on the Right to information of the people due to Section 44(3) of the DPDP Act, 2023. Thus, Statements I and II are correct and Statement II is the correct explanation of Statement I.

Q 69. B Andhau Inscriptions

- The Andhau Inscriptions are significant stone records from Kutch, India, primarily from the time of the Western Kshatrpa rulers Chashṭana and Rudradāman, dated to the Saka year 52 (130 CE), detailing memorial pillars (chhayastambhas) for individuals like Rishabhadeva, mentioning key genealogical figures (Chashṭana, Jayadaman, Rudradaman) and reflecting early Brahmanical memorial practices and the Kshatrpa lineage, with inscriptions found on hard stone slabs and later studied by scholars like R.D. Banerji.
- Published in Epigraphia Indica Vol. XVI, they consist of two stone inscriptions found at Andhau. They are primarily donative or memorial in nature.
- They confirm the rule of Rudradaman I in the Kutch region and provide insights into the administrative, social, and linguistic conditions of the 2nd century CE.
- These records are essential for understanding the history of the Saka (Western Kshatrpa) dynasty in Western India, acting as a primary source for the period immediately preceding the famous Junagadh rock inscription.
- Hence, option (b) is the correct answer.

- Q 70. B**
- The Bibighar massacre was a significant and tragic event that occurred in Kanpur (then Cawnpore) during the Indian Rebellion of 1857. On July 15, 1857, approximately 200 British women and children—who had survived an earlier ambush at Satichaura Ghat—were executed by rebel forces. The bodies of the victims were subsequently thrown into a nearby dry well.
 - The Massacre Ghat is located on the southern bank of the River Ganges and marks the northern boundary of the city of Kanpur, and became historically important during the Indian Rebellion of 1857 (the Indian Sepoy Mutiny). Known as the Siege of Cawnpore (now Kanpur) in British colonial records, took place on June 27, 1857. When 300 British men, women and children were slaughtered, and later 120 women and children were taken to Kanpur, killed and thrown into a well (known as the Bibighar Massacre). The murders greatly embittered the British rank-and-file against the Sepoy rebels and inspired the war cry "Remember Cawnpore".
 - Hence, option (b) is the correct answer.

- Q 71. D**
- Net Primary Productivity (NPP) is the energy (biomass) remaining after plants use part of their Gross Primary Productivity (GPP) for respiration,, making it the actual food available to primary consumers, and is generally highest in tropical rainforests due to favorable, year-round conditions.

Primary producers (like plants) convert solar energy into chemical energy through photosynthesis. Some of this energy is used by the plant for its own survival (respiration). The energy that remains—stored as plant tissue or biomass—is what is actually available for herbivores (primary consumers) to eat. **Hence, statement 1 is correct.**

Due to high temperatures, high rainfall, and consistent sunlight, tropical rainforests exhibit the highest NPP among terrestrial ecosystems, surpassing temperate forests. **Hence, statement 2 is correct.**

- GPP (Gross Primary Productivity): The total amount of energy captured by autotrophs.
- R (Respiration): The energy the plants consume for their own metabolic needs.
- NPP: The "net" gain that stays in the system as growth. **Hence, statement 3 is correct.**
- **$NPP = GPP - R$**

Q 72. C Though the phrase 'Judicial Review' has nowhere been used in the Constitution, the provisions of several Articles explicitly confer the power of judicial review on the Supreme Court and the High Courts. Some of the provisions are explained below:

- Article 13 declares that all laws that are inconsistent with or in derogation of the Fundamental Rights shall be null and void.
- Article 32 guarantees the right to move the Supreme Court for the enforcement of the Fundamental Rights and empowers the Supreme Court to issue directions or orders or writs for that purpose.
- Article 135 empowers the Supreme Court to exercise the jurisdiction and powers of the Federal Court under any preconstitution law.
- **Article 136 authorises the Supreme Court to grant special leave to appeal from any court or tribunal (except military tribunal and court martial).**
- **Article 226 empowers the High Courts to issue directions or orders or writs for the enforcement of the Fundamental Rights and for any other purpose.**
- **Article 227 vests in the High Courts the power of superintendence over all courts and tribunals within their respective territorial jurisdictions (except military courts or tribunals).**
- Article 246 deals with the subject matter of laws made by Parliament and by the Legislatures of States (i.e., Union List, State List and Concurrent List).
- **Hence, option (c) is the correct answer.**

Q 73. D Context: Recently, the World Inequality Lab released the World Inequality Report 2026.

World Inequality Report 2026.

The World Inequality Report (WIR) 2026 is the third major edition (following 2018 and 2022) released by the World Inequality Lab (WIL), a research centre based at the Paris School of Economics.

Key Findings of the Report

Global Wealth Concentration: Top 1% control 37% and top 10% share 75% of all global wealth.

- India: Top 1% holds about 40% and top 10% hold around 65% of total wealth.

Global Income Concentration: Top 10% share 53% of global income.

- India: Top 1% earns around 23% and top 10% capture 58% of national income.
- **Wealth Growth:** Since the 1990s, the wealth of billionaires has grown (at 8%) nearly twice the rate of bottom half of the population.
- **Climate Inequality:** Top 10% account for 77% of carbon emissions, while the poorest half accounts for only 3%.

- Structural Imbalance: Each year Global South transfer more than 1% of world GDP to Global North through debt service, profit repatriation, and financial flows (3 times more than development aid).

Hence, option (d) is the correct answer.

- Q 74. B**
- **Article 348(1)(a) states that all proceedings in the Supreme Court and every High Court shall be in the English language unless Parliament by law otherwise provides.** This ensures uniformity in legal proceedings across India. **Hence, statement 1 is correct.**
 - **Article 348(2) allows the Governor of a state, with the permission of the President, to authorize the use of Hindi or any other official language of the state in the proceedings of the High Court.** However, judgments, decrees, and orders must still be in English unless Parliament provides otherwise. **Hence, statement 2 is not correct.**
 - **Article 348(1)(b) states that the authoritative text of all Bills, Acts, Ordinances, Orders, Rules, and Regulations at both the Union and State levels shall be in English until Parliament provides otherwise by law.** This ensures clarity and consistency in legal documents. **Hence, statement 3 is correct.**
 - **Article 348(3) allows State Legislatures to adopt Hindi or any other official language for official purposes within the state.**
 - **Hence, option(b) is the correct answer.**

- Q 75. A**
- **In 2002, for the first time, the Government of India introduced a Cooperation Policy, but now proving to be outdated due to the radical shifts brought on by globalization, digitization, and socio-economic transformation. Hence, statement 1 is not correct.**
 - **Vision:** To significantly contribute to India's collective ambition of becoming 'Viksit' by 2047 through the promotion of an environment conducive to sustainable cooperative development, following the vision of 'Sahkar-se-Samridhi'.
 - By 2034, the National Cooperation Policy aims to triple the cooperative sector's contribution to GDP, bring 50 crore active members into the fold, and connect youth with employment opportunities. **Hence, statement 2 is not correct.**
 - There will be at least one primary cooperative unit in every panchayat, which could be a Primary Agricultural Credit Society (PACS), Primary Dairy Cooperative, Primary Fisheries Cooperative, Primary Multipurpose PACS, or any other primary unit.
 - 30 per cent increase in the number of cooperative societies and the establishment of at least one cooperative society in every village are key objectives of the New Cooperation Policy
 - **A robust multi-tier implementation structure is proposed:**
 - Implementation Cell within the Ministry of Cooperation with technical Project Management Unit support for effective and timely implementation of the policy.
 - **National Steering Committee on Cooperation Policy chaired by the Union Cooperation Minister will be constituted for overall guidance, inter-ministerial coordination, periodic policy review, etc. Hence, statement 3 is correct.**
 - Policy Implementation and Monitoring Committee headed by the Union Cooperation Secretary for coordination with States, troubleshooting implementation bottlenecks, periodic monitoring and evaluation, etc.

- Q 76. A** The Barabar Caves are a group of **ancient rock-cut caves located in the Jehanabad district of Bihar, India**. They are believed to date back to the Maurya period (circa 322–185 BCE) and are among the earliest examples of rock-cut architecture in India. **Hence statement 1 is not correct.**
- **Age and Origin:** The caves are attributed to Emperor Ashoka, who had them carved for the Ajivikas, a religious sect during the Maurya dynasty.
 - **Architecture:** The caves are known for their polished, well-finished surfaces. The **architectural style is indicative of the transition from wooden architecture to stone architecture** during that period. The caves exhibit unique architectural features, including the curved shape of the interiors. The polished surfaces inside the caves are remarkable.
 - **Cave Complex:** The **main caves at Barabar are Lomas Rishi Cave, Sudama Cave, and Karan Chaupar Cave**. The nearby Nagarjuna Hills also house additional caves, including Gopi Cave and Vapiya-ka-Kubha.
 - **Inscriptions:** Inscriptions in the caves provide historical information and attribute them to Ashoka, **highlighting his patronage of diverse religious beliefs and practices. They reflect the religious tolerance and inclusivity promoted by Emperor Ashoka. Hence statement 2 is correct.**
 - **Cultural Heritage:** Barabar Caves are a protected archaeological site and are designated as a National Monument. They **are included in the UNESCO World Heritage tentative list. Hence statement 3 is not correct.**

Hence option (a) is the correct answer.

- Q 77. A** Under India's Plastic Waste Management (Amendment) Rules, PET (Polyethylene Terephthalate) bottles are classified as Category I: Rigid Plastic Packaging. Producers, Importers, and Brand Owners (PIBOs) are legally responsible for the collection and recycling of the plastic packaging they introduce into the market. **Hence, statement 1 is correct.**

In 2022, the Food Safety and Standards Authority of India (FSSAI) overturned a previous ban and issued guidelines allowing the use of recycled PET (rPET) for food-contact materials, provided it meets specific safety standards and is processed using FSSAI-approved technologies. Further amendments in 2025 have reinforced these standards to support a circular economy. **Hence, statement 2 is correct.**

While PET is highly recyclable and many PET items (like large water bottles) are not on the "banned" list, they are not "completely exempted" from restrictions. For example, certain states have banned small-volume PET water bottles (e.g., bottles less than 200ml or 500ml). Furthermore, all PET packaging is strictly regulated under EPR mandates; "exempted" would imply they are outside the scope of restrictive regulations, which is not the case. **Hence, statement 3 is not correct.**

- Q 78. C** The doctrine of 'sovereignty of Parliament' is associated with the British Parliament. Sovereignty means the supreme power within the State. That supreme power in Great Britain lies with the Parliament. There are no 'legal' restrictions on its authority and jurisdiction. The Indian Parliament, on the other hand, cannot be regarded as a sovereign body in the similar sense as there are 'legal' restrictions on its authority and jurisdiction. The factors that limit the sovereignty of Indian Parliament are:
- **Written Nature of the Constitution:** The Constitution is the fundamental law of the land in our country. It has defined the authority and jurisdiction of all the three organs of the Union government and the nature of interrelationship between them. Hence, the Parliament has to operate within the limits prescribed by the Constitution. There is also a legal distinction between

the legislative authority and the constituent authority of the Parliament. Moreover, to effect certain amendments to the Constitution, the ratification of half of the states is also required. In Britain, on the other hand, the Constitution is neither written nor there is anything like a fundamental law of the land.

- **Federal System of Government:** India has a federal system of government with a constitutional division of powers between the Union and the states. Both have to operate within the spheres allotted to them. Hence, the lawmaking authority of the Parliament gets confined to the subjects enumerated in the Union List and Concurrent List and does not extend to the subjects enumerated in the State List (except in five abnormal circumstances and that too for a short period). Britain, on the other hand, has a unitary system of government and hence, all the powers are vested in the Centre.
- **System of Judicial Review:** The adoption of an independent Judiciary with the power of judicial review also restricts the supremacy of our Parliament. Both the Supreme Court and high courts can declare the laws enacted by the Parliament as void and ultra vires (unconstitutional), if they contravene any provision of the Constitution. On the other hand, there is no system of judicial review in Britain. The British Courts have to apply the Parliamentary laws to specific cases, without examining their constitutionality, legality or reasonableness.
- **Fundamental Rights:** The authority of the Parliament is also restricted by the incorporation of a code of justiciable fundamental rights under Part III of the Constitution. Article 13 prohibits the State from making a law that either takes away totally or abrogates in part a fundamental right. Hence, a Parliamentary law that contravenes the fundamental rights shall be void. In Britain, on the other hand, there is no codification of justiciable fundamental rights in the Constitution. The British Parliament has also not made any law that lays down the fundamental rights of the citizens. However, it does not mean that the British citizens do not have rights. Though there is no charter guaranteeing rights, there is maximum liberty in Britain due to the existence of the Rule of Law.
- **Hence, option (c) is the correct answer.**

- Q 79. C**
- The Supreme Court of India comprises the Chief Justice and 33 other Judges appointed by the President of India. Supreme Court Judges retire upon attaining the age of 65 years. In order to be appointed as a Judge of the Supreme Court, a person must be a citizen of India and must have been, for at least five years, a Judge of a High Court or of two or more such Courts in succession, or an Advocate of a High Court or of two or more such Courts in succession for at least 10 years or **he must be, in the opinion of the President, a distinguished jurist. Hence option (a) is not correct.**
 - Judges of the Federal Court in the USA are appointed by the President with the advice and consent of the Senate. The candidates are assessed by a committee of the American Bar Association and reviewed by the Senate Judiciary Committee before a vote in the Senate. **There is no set retirement age for judges in the US as they continue to hold office for "good behaviour". In India, a judge of Supreme Court though does not have a fixed tenure, he/she must vacate the seat after attaining the age of 65 years. Hence, the Supreme court judge in India is not a judge for life. Hence option (b) is not correct.**
 - **A judge of the Supreme Court can be removed from his Office by an order of the president. The President can issue the removal order only after an address by Parliament has been presented to him in the same session for such removal. The address must be supported by a special majority of each House of Parliament (ie, a majority of the total membership of that**

House and a majority of not less than two-thirds of the members of that House present and voting). The grounds of removal are two—proved misbehaviour or incapacity. Hence, option (c) is the correct answer.

- When there is a lack of quorum of the permanent judges to hold or continue any session of the Supreme Court, **the Chief Justice of India can appoint a judge of a High Court as an ad hoc judge of the Supreme Court for a temporary period.** He can do so only after consultation with the chief justice of the High Court concerned and with the previous consent of the president. The judge so appointed should be qualified for appointment as a judge of the Supreme Court. It is the duty of the judge so appointed to attend the sittings of the Supreme Court, in priority to other duties of his office. While so attending, he enjoys all the jurisdiction, powers and privileges (and discharges the duties) of a judge of the Supreme Court. **Hence option (d) is not correct.**

Q 80. A • Subhash Chandra Bose Aapda Prabandhan Puraskar-2025, was given to the Indian National Centre for Ocean Information Services (INCOIS) in the Institutional Category, for its excellent work in Disaster Management. Hence, option (a) is the correct answer.

- **About the Subhash Chandra Bose Aapda Prabandhan Puraskar:**
 - It is an annual award to recognize and honour the invaluable contribution and selfless service rendered by individuals and organizations in India in the field of disaster management.
 - The award is announced every year on 23rd January, the birth anniversary of Netaji Subhash Chandra Bose.
 - The award carries a cash prize of Rs. 51 lakh and a certificate in case of an institution and Rs. 5 lakh and a certificate in case of an individual.

Q 81. C • Yogacara or Vijnanavadin School is one of the chief philosophical schools of the Great Vehicle i.e. Mahayana Buddhism. It is an influential tradition of Buddhist philosophy emphasizing the study of cognition, perception, and consciousness through the interior lens of meditative and yogic practices.

- Yogacara is the school which follows idealism and reject realism or qualified realism. **The world according to them is built by consciousness and is no more than a dream. The only reality is “Suchness”.**
- Another School which comes from **Great Vehicle is Madhyamika. It takes the middle line between uncompromising realism and idealism and thus called so.**
- This School is also called Sunyavada as it believes in **theory of Void or Emptiness.** According to them, the world which is seen as flux of interdependent events by most schools is unreal and so as the consciousness which perceived it so. Therefore Sansara doesn't exist and therefore there is no Nirvana also. There is no difference between Sansara and Nirvana and they are same non-entity. The only thing that exist is the Void or Emptiness.
- **Nagarjuna, contemporary of Kaniska, is one of the great philosophers of Madhyamika School.** Bhaviveka, Chandrakriti, Santaraksita are some more philosophers of Madhyamika School.
- **Asanga and Vasubandhu along with great logicians Dignaga and Dharmakirti are great philosophers of Yogacara School.**
- Madhyamika Karika by Nagarjuna forms the basic text of the Madhyamika School. **The most important writings of Yogacara is the Lankavatara Sutra & Sutralankara.**

- Hence both the statements are correct.

- Q 82. C**
- Government extends price support to wheat and paddy through FCI and State Agencies. Government agencies undertake MSP operation at mandis/ temporary purchase centres/aggregation points. Location and number of purchase centres to be opened are decided in consultation with /by the State governments.
 - **Statement 1 is correct: Under centralized procurement system, the procurement of foodgrains (wheat and rice) in Central Pool is undertaken either by FCI directly or by State Govt. Agencies (SGA).** Quantity procured by SGAs is handed over to FCI for storage and subsequent issue against GOI allocations in the same State or movement of surplus stocks to other States. The cost of the foodgrains procured by State agencies is reimbursed by FCI as per Provisional per cost-sheet issued by GOI as soon as the stocks are delivered to FCI.
 - **Under Decentralized (DCP) Procurement system, the State Government/ its agencies procure, store and distribute rice /wheat/coarse grains within the state.** The excess stocks (Rice & wheat) procured by the State /its agencies are handed over to FCI in Central Pool. The expenditure incurred by the State Government on procurement, storage and distribution of DCP stocks are reimbursed by Government of India.
 - **Statement 2 is correct: The expenses such as MSP, arhatiya/society commission, administrative charges, mandi labour charges, transportation charges, custody & maintenance charges, interest charges, gunny cost, milling charges and statutory taxes are reimbursed on actual basis.** The cost of excess stocks handed over to FCI is reimbursed by FCI to the State Government/agencies as per Government of India costs sheet.
 - Procurement at MSP is open ended i.e., whatever foodgrains are offered by the farmers ,within the stipulated procurement period and which conforms to the quality specifications prescribed by Government of India, are purchased at MSP (and bonus/incentive ,if any) by the Government agencies including FCI for central Pool.

- Q 83. D**
- Hoysala temple architecture is a style that originated during the Hoysala Empire and flourished between the 11th and 14th centuries in the present-day Indian state of Karnataka.

Key features of Hoysala Temple Architecture:

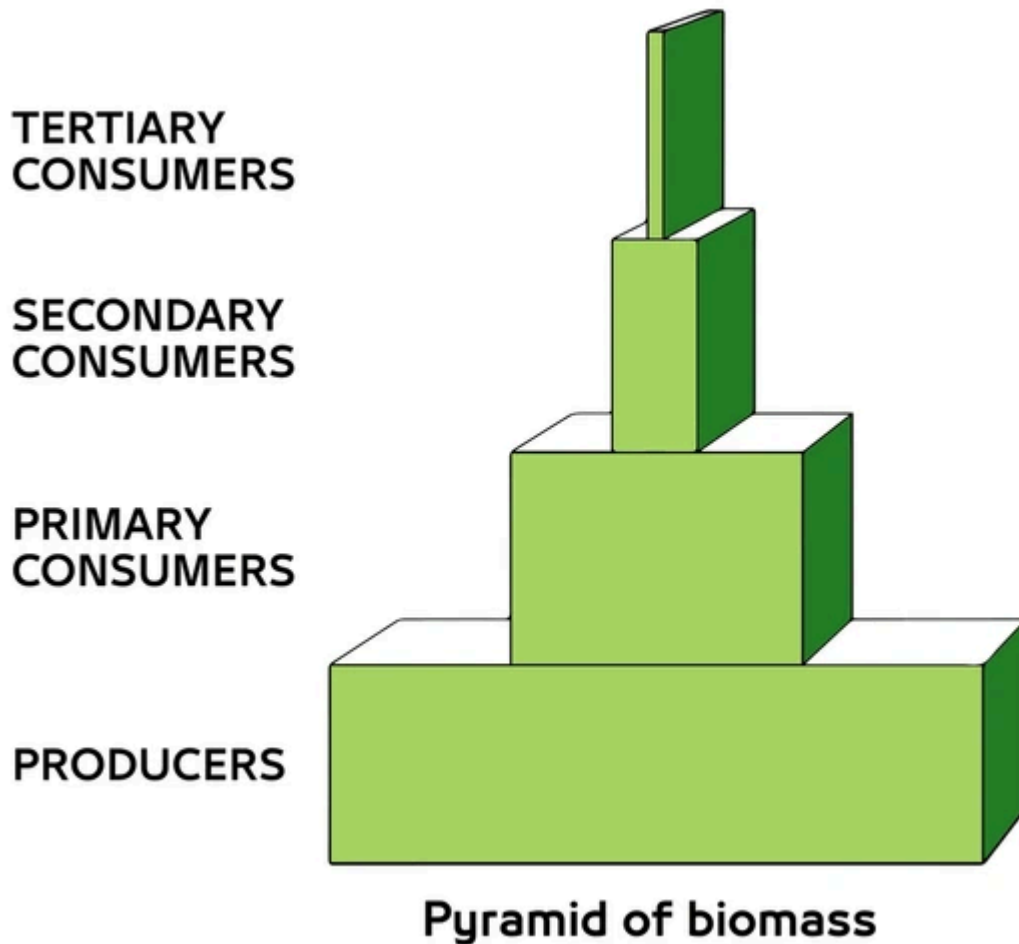
- **Stellate Ground Plan:** Hoysala temples are renowned for their star-shaped layouts. This shape enhances visual complexity and allows for ornate decorations. Hence, option 1 is correct.
- **Material:** Built mainly from soapstone, the soft stone permits detailed carvings and sculptures.
- **Intricate Carvings:** Temples feature elaborate sculptures depicting deities, mythological scenes, and decorative motifs, showcasing the craftsmanship of local artisans.

Architectural Elements:

- **Jagati:** Many temples are built on raised platforms that serve as paths for circumambulation. Hence option 2 is correct.
- **Decorative Pillars:** Lathe-turned pillars adorned with carvings support the temple structures.
- **Ornate Ceilings:** Ceilings often feature domical shapes with intricate designs.
- **The walls and stairs of the temple followed a zigzag pattern. Hence option 3 is correct.**
- **Notable Examples:** Chennakesava Temple in Belur and Hoysaleswara Temple in Halebidu

Q 84. A In a terrestrial ecosystem, the primary producers (first trophic level) contain the highest biomass. This base level, consisting of green plants, trees, and grasses, supports the entire food chain, with biomass decreasing significantly at each successive trophic level—herbivores (primary consumers) and then carnivores (secondary/tertiary consumers).

Upright Pyramid: Terrestrial ecosystems feature an upright biomass pyramid, where the base (producers) is the widest.



Hence, option (a) is the correct answer.

- Q 85. C**
- Article 14 of the Indian Constitution guarantees that no person shall be denied the right to equality before law or the equal protection of law in the territory of India. This is a right that can be claimed by any person, whether a citizen or a non-citizen, on Indian soil. Here, we can find that Article 14 comprises two expressions, which are 'equality before law' and 'equal protection of law'. The first expression 'equality before law' is borrowed from the English common law. The expression 'equal protection of law' is borrowed from the Constitution of the United States of America. Section 4 of the 14th Amendment to the American Constitution states that no person shall be denied the equal protection of law by any state within its jurisdiction.
 - The expression 'equal protection of law' is a positive one unlike 'equality before law'. It simply means that all persons in similar circumstances shall be given the same rights and liabilities. It essentially means that equals are to be treated equally and there must be no discrimination amongst them. Equals and unequal cannot be put in the same footing and be treated without discrimination.

Hence option (c) is the correct answer.

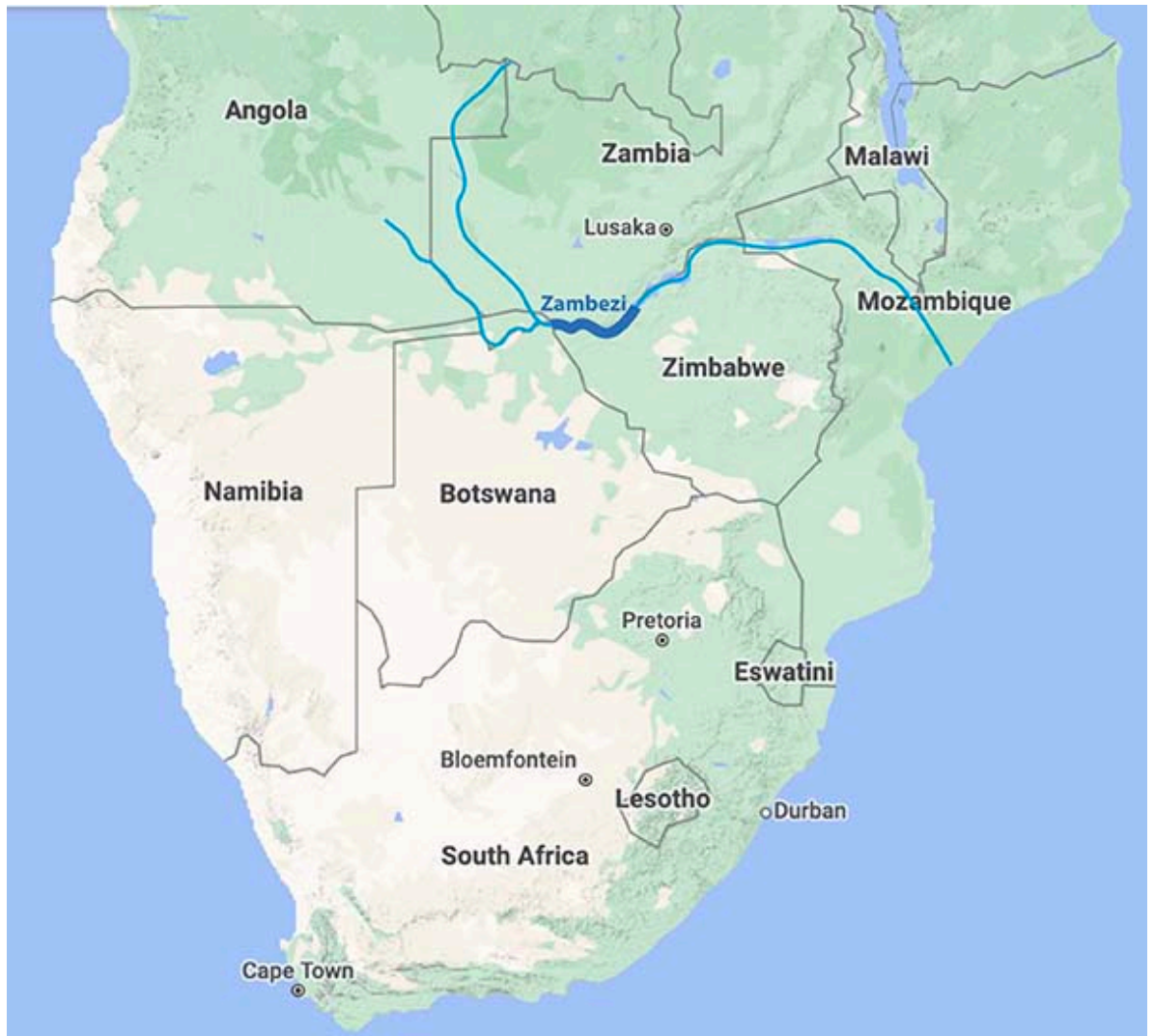
- Q 86. A • Nile:** It flows North across the African continent and drains into the Mediterranean Sea. It is the longest river in the world, but it does not reach the Indian Ocean.



- **Mekong:** It flows through Southeast Asia (China, Myanmar, Laos, Thailand, Cambodia, and Vietnam) and drains into the South China Sea (Pacific Ocean).



- **Zambezi:** This river flows through southern Africa and drains into the Indian Ocean. It famously flows through the Victoria Falls and enters the ocean in Mozambique.



- **Congo:** It is the deepest river in the world and flows through Central Africa. It drains into the Atlantic Ocean.



- Hence, option (a) is the correct answer.

Q 87. B 1. Kanchenjunga: Located in India (and Nepal)

- Location: Situated on the border between Sikkim (India) and Nepal.
- Significance: It is the third-highest mountain in the world and the highest peak located in India (considering K2 is in POK). Because a significant portion of its massif lies within the Indian state of Sikkim, it is counted as being located in India.

2. Dhaulagiri: Located in Nepal

- Location: Situated entirely within the borders of Nepal.
- Significance: It is the seventh-highest mountain in the world. It is part of the Dhaulagiri massif in the Himalayas of north-central Nepal.

3. Kamet: Located in India

- Location: Situated in the Chamoli district of Uttarakhand, India.
- Significance: It is the second-highest mountain in the Garhwal region of Uttarakhand, after Nanda Devi. It is located near the Tibetan border but is entirely within Indian territory.
-  Kamet on map of india - Brainly.in

4. Annapurna: Located in Nepal

- Location: Situated entirely within Nepal.
- Significance: It is a massif in the Himalayas of north-central Nepal that includes one peak over 8,000 meters.



- Hence, option (b) is the correct answer.

Q 88. D • In an open vessel, water boils at approximately 100°C (at sea level). This is because the vapor pressure of the water equals the atmospheric pressure. Once water reaches its boiling point, its temperature stays constant at 100°C, regardless of how much heat you add; the extra energy simply turns the liquid into steam.

Pressure Cooker Working:

- A pressure cooker is a sealed environment. As the water inside is heated, the steam produced cannot escape. This trapped steam increases the internal pressure inside the cooker. As the pressure increases, it becomes harder for water molecules to escape the liquid phase and turn into

gas. Consequently, the boiling point of the water rises, often reaching temperatures around 120°C.

- Cooking is essentially a series of chemical reactions (like the breaking down of fibers or denaturing proteins). Most chemical reactions happen faster at higher temperatures. **Because the water in a pressure cooker is at 120°C instead of 100°C, it transfers more thermal energy to the food per second, significantly reducing the total cooking time. Hence, option (d) is the correct answer.**

Q 89. A • Special Economic Zone (SEZ) is a territory within a country that is typically duty-free (Fiscal Concession) and has different business and commercial laws chiefly to encourage investment and create employment i.e **a designated duty free enclave to be treated as a territory outside the customs territory of India for the purpose of authorised operations in the SEZ.**

India was one of the first in Asia to recognize the effectiveness of the Export Processing Zone (EPZ) model in promoting exports, with **Asia's first EPZ set up in Kandla in 1965. Hence statement 1 is not correct.**

- **While these EPZs had a similar structure to SEZs**, the government began to establish SEZs in 2000 under the Foreign Trade Policy to redress the infrastructural and bureaucratic challenges that were seen to have limited the success of EPZs.

The Special Economic Zones Act was passed in 2005. The Act came into force along with the SEZ Rules in 2006.

- However, SEZs were operational in India from 2000 to 2006 (under the Foreign Trade Policy).
- The SEZ Rules provide for different minimum land requirement for different class of SEZs. Every SEZ is divided into a processing area where alone the SEZ units would come up and the non-processing area where the supporting infrastructure is to be created.

Some incentives for setting up a sourcing or manufacturing platform within an Indian SEZ include:

- Duty free import and domestic procurement of goods for the development, operation, and maintenance of your company/SEZ unit; **Hence statement 2 is correct.**
- 100 percent income tax exemption on export income for first five years, 50 percent for five years thereafter, and 50 percent of the export profit reinvested in the business for the next five years (Sunset Clause for Units will become effective from April 1, 2020);
- **Exemption from the goods and services tax (GST) and levies imposed by state government (supplies to SEZs are zero rated under the IGST Act, 2017, meaning they are not taxed); Hence statement 3 is correct.**
- **No licence required for import;**
- Single window clearances for all state and federal government approvals;
- Exemption in electricity duty and tax on sale of electricity by certain states in India;
- Presence of customs officer in the SEZs to facilitate and expedite the trade processes; and
- Some states also offer land to SEZ developers at concessional rates to promote industries in accordance with the state's prevailing Industrial Policy.

Q 90. C • Peatlands are waterlogged wetland ecosystems defined by the accumulation of partially decomposed organic matter (peat) over centuries. Covering only ~3% of Earth's land, they are the world's largest natural terrestrial carbon store, holding twice as much carbon as all forests

combined. They are critical for climate regulation, biodiversity, and water management. **Hence, Assertion A is correct.**

- Peat accumulation occurs precisely because of slow (not fast) decomposition, driven by waterlogged, anaerobic (oxygen-poor), and acidic conditions. While tropical peatlands exist, they form through inhibited decomposition, not fast decomposition. **Hence, Reason R is not correct.**
- **Hence, option (c) is the correct answer.**

- Q 91. B**
- The Doomsday Clock is a metaphor maintained by the Bulletin of the Atomic Scientists. Since its creation in 1947, it has served as a symbolic gauge rather than a literal or scientific measurement tool.
 - "Midnight" on the clock represents a human-made global catastrophe that could potentially end civilization. Experts adjust the clock's hands based on evolving threats, primarily focusing on nuclear risks, climate change, and disruptive technologies like artificial intelligence.
 - As of early 2026, the clock is set at 85 seconds to midnight, the closest it has ever been to the brink.
 - **Hence, option (b) is the correct answer.**

- Q 92. A**
- Article 311 places two restrictions on the above 'doctrine of pleasure'. In other words, it provides two safeguards to civil servants against any arbitrary dismissal from their posts:
- **A civil servant cannot be dismissed or removed by an authority subordinate to that by which he was appointed. Hence, statement 1 is correct.**
 - A civil servant cannot be dismissed or removed or reduced in rank except after an inquiry in which he has been informed of the charges against him and given a reasonable opportunity of being heard in respect of those charges.
- However, the second safeguard (holding inquiry) is not available in the following three cases:**
- Where a civil servant is dismissed or removed or reduced in rank on the ground of conduct which has led to his conviction on a criminal charge; or **(Hence statement 2 is not correct).**
 - **Where the authority empowered to dismiss or remove a civil servant or to reduce him in rank is satisfied that for some reason (to be recorded in writing), it is not reasonably practicable to hold such inquiry; or**
 - **Where the president or the governor is satisfied that in the interest of the security of the state, it is not expedient to hold such an inquiry. Hence, option (a) is the correct answer.**

- Q 93. C**
- From c. sixth century CE onwards finds of gold coins taper off.
 - Some suggest that with the **collapse of the Western Roman Empire** long-distance trade declined, and this **affected the prosperity** of the states, communities and regions that had benefited from it. **Hence statements 1 and 2 are correct.**
 - Others argue that **new towns and networks of trade started emerging** around this time. **Hence statement 3 is not correct.**
 - They also point out that though finds of coins of that time are fewer, coins continue to be mentioned in inscriptions and texts.

- Q 94. A**
- Invasive species are non-native organisms (plants, animals, pathogens) introduced to new environments, intentionally or accidentally, that cause significant ecological, economic, or human

health harm. They spread rapidly, outcompete native species, and alter ecosystems, driven by high reproduction rates, adaptability, and lack of natural predators.

Characteristics increase invasiveness of a species

- High reproductive rate: Invasive species are often "r-strategists," meaning they produce a large number of offspring in a short period. This allows them to establish a population quickly and outcompete native species that may have slower reproduction cycles. **Hence, point 1 is correct.**
- Broad ecological tolerance: To survive in a new, unfamiliar environment, a species needs to be a "generalist." Species with a broad ecological tolerance can withstand various temperatures, moisture levels, and food sources, making it easier for them to adapt to diverse habitats. **Hence, point 2 is correct.**
- Absence of natural predators: In their native habitats, species are usually kept in check by specific predators, diseases, or parasites. When introduced to a new area where these natural enemies are missing, their population can grow unchecked, leading to an "explosion" in numbers. **Hence, point 4 is correct.**

Specialized niche requirement is a characteristic of specialist species, which are typically more vulnerable to extinction because they depend on specific resources or conditions. In contrast, invasive species are usually generalists with broad niches that allow them to adapt easily to new or disturbed habitats. **Hence, point 3 is not correct.**

Q 95. C About Biological Weapon Convention (BWC)

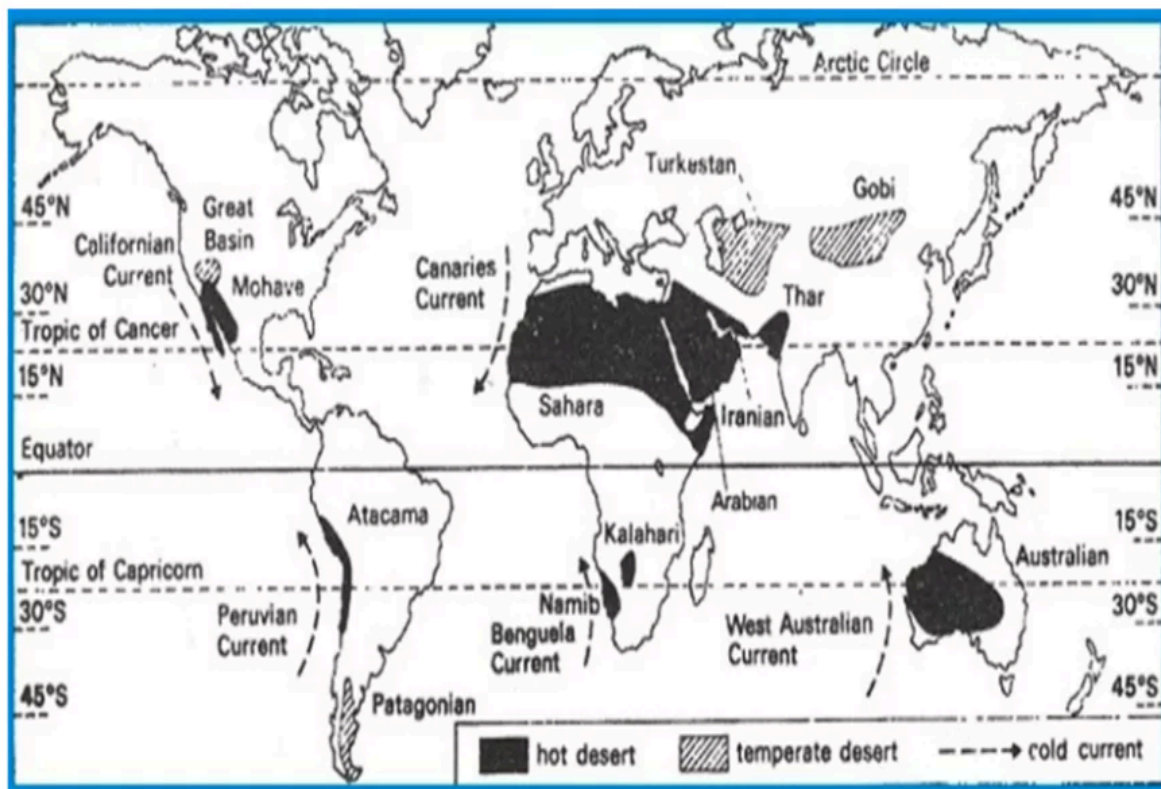
- Formally known as 'The Convention on the Prohibition of the Development, Production and Stockpiling of Bacteriological (Biological) and Toxin Weapons and on their Destruction'.

Nature: It was the first multilateral disarmament treaty to ban an entire class of weapons of mass destruction (WMDs). Hence, statement 1 is correct.

- BWC supplements the 1925 Geneva Protocol, which had prohibited only the use of biological weapons.
- **Establishment:** Opened for signature on 10 April 1972 and entered into force on 26 March 1975.
- **Membership:** 189 States Parties and four Signatory States, India signed and ratified in 1974.
- Under Article I of the BWC, the prohibition applies to biological agents and toxins "of types and in quantities that have no justification for prophylactic, protective or other peaceful purposes". This means the use of these agents for prophylactic purposes (such as the development of vaccines or protective medical research) is explicitly not prohibited and is considered a legitimate, peaceful application. **Hence, statement 2 is correct.**

- Q 96. B**
- Western Disturbances are weather systems originating in the Mediterranean region. However, they are a winter phenomenon (occurring primarily from December to March) and bring much-needed, though sparse, rainfall to Northwest India.
 - They do not occur frequently during the summer, nor is their "presence" a reason for the desert's aridity. The lack of rainfall in summer is due to the failure of the Monsoon branches, not the interference of Western Disturbances. **Hence, statement 1 is not correct.**
 - The Aravalli Range is situated in a South-West to North-East direction. The Arabian Sea branch of the Southwest Monsoon blows in the same direction.

- Because the mountains are parallel to the wind direction, they do not provide an orographic barrier. The moisture-laden winds simply pass by without being forced to rise, cool, and condense into rain. **Hence, statement 2 is correct.**
- The Thar Desert is located approximately between 25° N and 30° N latitude, placing it directly under the Subtropical High-Pressure Belt. This area is characterized by the descending limb of the Hadley Cell. As air sinks (subsides), it compresses and warms up, which increases its moisture-holding capacity and prevents cloud formation. This atmospheric stability is a major global reason for the existence of subtropical deserts. **Hence, statement 3 is correct.**



Q 97. B Context of the Period

- By the late 1930s, Indian politics was witnessing heightened debates over participation in the Second World War. The British unilaterally declared India a participant in the war in 1939 without consulting Indian leaders. This created a divide within the Congress and intensified nationalist opposition.

Emergence of the Forward Bloc

- Subhas Chandra Bose, after resigning from the Congress presidency in 1939 due to differences with Gandhi and the Congress High Command, founded the Forward Bloc. Its objective was to consolidate all leftist and radical elements under one platform and pursue uncompromising struggle against British imperialism.

The Anti-Compromise Conference, Ramgarh (March 1940)

- The Forward Bloc convened the Anti-Compromise Conference at Ramgarh soon after the Congress session was held at the same place.
- Its aim was to oppose any compromise with British rule and to mobilize mass resistance against India's forced involvement in the Imperialist War.
- Bose used the conference to highlight that Congress leaders were showing excessive restraint, while the hour demanded militant struggle.

- Hence, option (b) is the correct answer.

Significance

- It marked Subhas Bose's assertion as an independent nationalist leader, distancing himself from the moderate and Gandhian line within Congress.
- The event paved the way for his subsequent revolutionary activities, eventually leading to his escape from India in 1941 and formation of the Indian National Army (INA) later.
- Unlike the Congress session at Ramgarh (where Gandhi was authorized to launch limited satyagraha), Bose's conference projected an alternative, radical path.

- Q 98. A**
- Recently, Bulgaria and Romania joined the Schengen Area of Europe.
 - The **Schengen zone is the world's largest passport-free travel and borderless zone in Europe.** It was created in 1985 in the Schengen town in Luxembourg. It covers 23 European Union member states plus the four members of the European Free Trade Association – Iceland, Norway, Switzerland, and Liechtenstein. It allows people, goods, and services to travel freely amongst its member countries without having to pass identity checks at checkpoints.
 - **Eurozone:** The Eurozone refers to an economic and geographic region comprising European Union countries that use the Euro as their national currency
 - Hence option (a) is the correct answer.

- Q 99. C** Chinese Scientists created the first Pangenome of Asian rice with the help of 'PacBio 'high-fidelity' (HiFi) Sequencing Technology is a single-molecule, real-time sequencing technology (SMRT) that provides incredible single-molecule read accuracy across long reads.

What is a Pangenome?

It is a collection of genome sequences from many individuals of the same species.

A genome is the complete set of genetic information in an organism.

- It is a kind of reference genome.
- Pangenome has the common genes as well as maps out the unique genes found in the individual varieties while, reference genomes generally confine themselves to identifying the characteristic genes that make up a species.

'high-fidelity' (HiFi) Sequencing Technology

- It is a single-molecule, real-time sequencing technology (SMRT) that provides incredible single-molecule read accuracy across long reads.
- It is a kind of Long-read sequencing method of genome sequencing.
- Long-read sequencing is a DNA (Deoxyribonucleic acid) sequencing approach that enables the sequencing of much longer DNA fragments than traditional short-read sequencing methods.

Key Applications:

- Agriculture: Helps in introducing new traits for disease-tolerance as well as resilience in crops against climate shocks.
- Precision Medicine: In humans, helps in identifying population-specific genetic markers for diseases. E.g. Human Pangenome Project.

Hence, option (c) is the correct answer.

- Q 100. D**
- **A remittance is a sum of money sent by one party to another, most commonly referring to funds transferred by foreign workers to their home country.** Inward remittance is the receipt of foreign currency into a home bank account (e.g., from overseas salary or family), usually converted to local currency. Outward remittance is the transfer of funds from a home country to a foreign country.
 - The LRS was introduced by the Reserve Bank of India on **February 4, 2004 (and not in 1999)**, although it is regulated under the Foreign Exchange Management Act (FEMA), 1999. **Hence statement 1 is not correct.**
 - **LRS is specifically designed for outward remittances and not inward (sending money from India to abroad) by resident individuals. Hence statement 2 is not correct.**
 - **Under the current LRS regulations, Indian resident individuals, including minors, are allowed to freely remit up to USD 250,000 per financial year (April – March) for permitted current or capital account transactions. Hence statement 3 is correct.**

Copyright © by Vision IAS

All rights are reserved. No part of this document may be reproduced, stored in a retrieval system or transmitted in any form or by any means, electronic, mechanical, photocopying, recording or otherwise, without prior permission of Vision IAS.